MAR ATHANASIUS COLLEGE OF ENGINEERING

KOTHAMANGALAM

(Government Aided Autonomous Institution)

Department of Computer Applications

Main Project Report

AI BASED HOTEL KITCHEN CLEANLINESS MONITORING SYSTEM

Done by

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Reg No: MAC24MCA-2041

Under the guidance of

Prof Bista K V

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MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM
(Government Aided Autonomous Institution)

CERTIFICATE

AI BASED HOTEL KITCHEN CLEANLINESS MONITORING SYSTEM

Certified that this is the bonafide record of project work done by

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during the fourth semester, in partial fulfilment of requirements for
the award of the degree,

Master of Computer Applications

of

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Nandhu C Sivadas

Abstract

The increasing demand for stringent food safety and early detection of sanitary hazards has made automated kitchen hygiene monitoring a critical component in the modern hospitality industry. Traditional diagnostic methods rely on periodic manual inspections by health officers, which are often time-consuming, subjective, and prone to human oversight. To overcome these limitations, this project presents an AI-Based Hotel Kitchen Cleanliness Monitoring System, designed to automatically detect and classify various hygiene-related indicators and biological hazards from visual data.

The proposed system is built using the state-of-the-art YOLOv8 (You Only Look Once version 8) architecture, optimized for real-time object detection and localization. The model is specifically trained to recognize nine distinct classes: Personal Protective Equipment (PPE) compliance including apron, hairnet, and gloves their corresponding violations no_apron, no_hairnet, and no_gloves and critical biological hazards such as rat, cockroach, and lizard. To ensure high detection precision in fast-paced culinary environments, the system utilizes a specialized dataset of approximately 10,000 images, curated to reflect real-world kitchen conditions including varied lighting, clutter, and camera angles.

To further enhance model robustness and resolve class imbalances particularly for rare pest sightings an advanced data engineering strategy was implemented. This included the use of the Albumentations library to create a pipeline, which simulated grainy CCTV noise and motion blur, improving the mean Average Precision (mAP@50) for biological hazards from 0.38 to over 0.77. The overall system achieves a strong performance with an aggregate mAP@50 of 92.4%, demonstrating its high reliability for automated safety auditing.

The entire project is developed as an end-to-end web application using the Django framework and an SQLite database, facilitating a dual-actor ecosystem. Hotel administrators can proactively upload surveillance media for instant AI evaluation, while public guests can submit visual evidence of poor hygiene to trigger administrative reviews. Based on real-time inference, the system dynamically calculates a hygiene score and normalized rating (1-5 stars), automating the generation of professional, downloadable PDF Compliance Certificates or detailed Violation Reports complete with fine assessments.

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1 INTRODUCTION

In recent years, maintaining proper hygiene standards in the food and hospitality industry has become a major global concern. Kitchens are one of the most important areas in hotels and restaurants because they directly influence food quality, customer health, and business reputa tion. Poor kitchen hygiene can lead to food contamination, bacterial growth, and the spread of foodborne diseases, which can cause serious health problems. Beyond direct health con sequences, unhygienic conditions pose a severe threat to a business's reputation and financial stability. Furthermore, failed health inspections often result in heavy financial penalties, legal actions, or even temporary closures of the establishment. Therefore, maintaining proper hy giene conditions in commercial kitchens is essential not only for customer safety but also for regulatory compliance and long-term business sustainability.

Traditionally, kitchen hygiene is monitored through manual inspections performed by health inspectors or management staff. These inspections are usually conducted periodically and de pend heavily on human observation and experience. Although manual inspection methods are widely used, they have several limitations. They are time-consuming, require skilled personnel, and are often subjective in nature. Human inspectors may miss certain hygiene violations due to workload, fatigue, or limited inspection time. Manual auditing is not only labor-intensive but also highly susceptible to human bias and exhaustion. Additionally, manual inspections cannot provide continuous monitoring, which is very important in fast-moving kitchen environments where cleanliness conditions can change rapidly.

Furthermore, the most critical hygiene hazards are highly dynamic. Pests such as rodents and insects are elusive, often appearing during off-hours or blending into dark kitchen en vi-ronments, making periodic human inspections highly ineffective for pest control. Similarly, kitchen staff compliance with Personal Protective Equipment (PPE) such as consis tently wear-ing hairnets, aprons, and gloves can fluctuate throughout a single shift. Ensuring strict adher-ence to these safety protocols requires continuous, objective real-world surveillance rather than a one-time manual checklist.

Due to these limitations, there is an increasing demand for automated and intelligent monitoring systems that can continuously evaluate kitchen hygiene conditions. With the rapid advance ment of modern technologies such as Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Computer Vision, and Internet of Things (IoT), new solutions are emerging for automated monitoring. IoT-based systems are already used in many industries to monitor environmental factors such as temperature, humidity, smoke levels, and gas leakage. However, most existing systems focus mainly on environmental monitoring and do not address visual cleanliness factors such as dirty surfaces, unclean utensils, waste accumulation, improper food handling, or pest intrusions. Computer vision combined with deep learning techniques can effectively analyze visual data and detect hygiene-related issues automatically. Among modern object detection algorithms, YOLO (You Only Look Once) has gained signifi cant attention because of its high speed and accuracy. YOLO processes images in a single step, making it faster and more efficient compared to traditional multi-stage detection models. Due to its real-time detection capability, YOLO is widely used in applications such as surveillance systems, medical imaging, autonomous vehicles, and industrial automation. These features make YOLO highly suitable for real-time kitchen hygiene monitoring applications where fast and accurate detection is required.

This project proposes an AI-Based Hotel Kitchen Cleanliness Monitoring System that uses deep learning-based object detection to automatically evaluate kitchen cleanliness. Instead of merely classifying a general environment, the system utilizes a custom-trained YOLOv8 ar chitecture to simultaneously detect and localize specific compliance and hazard classes. These include critical PPE elements such as aprons, hairnets, and gloves, as well as severe biological hazards like rats, cockroaches, and lizards. Based on these real-time detections, the kitchen is categorized into hygiene levels such as Clean, Moderately Clean, and Dirty.

Beyond the core detection model, the system is engineered as a comprehensive web-based platform utilizing the Django framework. This architecture facilitates a dual-actor ecosystem. Hotel administrators can proactively upload CCTV surveillance media for automated hygiene auditing, while public guests can submit visual evidence of poor hygiene to trigger admin istrative reviews. Based on the real-time AI inference, the system dynamically calculates a hygiene score and automatically generates official PDF Compliance Certificates for clean en vironments, or detailed Violation Reports complete with fine assessments for non-compliant kitchens. This scalability transforms hygiene management from a reactive, periodic chore into a proactive, data-driven operational strategy, ultimately fostering a culture of accountability and unparalleled food safety.

The main objective of this project is to develop an automated and reliable hygiene monitoring solution that reduces dependency on manual inspections and improves food safety standards. By integrating AI, computer vision, and web technologies, the proposed system helps improve hygiene monitoring efficiency and supports smart automation in the hospitality industry. By leveraging the YOLOv8 object detection model, the system ensures that these threats are iden tified with high precision and minimal latency. Furthermore, the inclusion of an automated reporting mechanism allows administrators to generate transparent compliance certificates and detailed violation records seamlessly. Ultimately, the project bridges the gap between advanced computer vision research and practical, everyday commercial kitchen management.

2 SUPPORTING LITERATURE

In recent years, significant research has been conducted on the use of Artificial Intelligence (AI), Computer Vision, and Deep Learning techniques for maintaining hygiene and safety in restaurant and kitchen environments. These studies highlight the effectiveness of automated monitoring systems in reducing human error, improving hygiene compliance, and ensuring food safety. The following literature provides a strong foundation for the proposed AI-Based Hotel Kitchen Cleanliness Monitoring System.

2.1 Literature Review

PAPER 1

Nayar, A., et al. "An Automated Alert System for Monitoring the Hygiene in Restaurants Using Machine Vision." IEEE International Conference on Trends in Engineering Systems and Technologies (ICTEST), 2024.

Maintaining consistent hygiene standards in restaurant kitchens is a critical challenge due to the high speed of operations and the inherent limitations of manual periodic inspections. Traditionally, these environments rely on subjective human oversight, which is often compromised by fatigue and the inability to provide continuous monitoring. In this paper, the authors propose an end-to-end automated monitoring system that utilizes machine vision to enforce safety protocols. A key innovation in this work is the integration of a cloud-based alerting mechanism upon detecting a violation, the system automatically triggers an alert sent via a mobile application to administrative authorities, ensuring a proactive response to potential contamination risks. The research demonstrates a transition from passive video recording to an active, interventionist safety strategy that significantly reduces the administrative burden of manual auditing.

The system is built upon a custom-trained YOLOv8 (You Only Look Once version 8) architecture, chosen for its ability to perform real time object detection with high localization accuracy. The model is specifically trained to identify a variety of hygiene violations, including improper staff attire such as missing aprons, gloves, or hairnets and the presence of critical biological hazards like rats and cockroaches. The research emphasizes that the utilization of YOLOv8's anchor free detection mechanism allows for higher precision in identifying small, fast-moving objects against complex kitchen backgrounds. By leveraging high-speed inference, the system ensures potential threats are identified with minimal latency even under the grainy or low-light conditions typical of standard CCTV feeds.

To ensure high detection precision in fast-paced culinary environments, the system utilizes a specialized dataset of 4,000 images. This dataset is categorized into two primary domains: Attire Compliance (3,000 images) and Biological Hazards (1,000 images). The study details the importance of a diverse training dataset, noting that the inclusion of varied lighting conditions and camera angles is vital for maintaining model robustness in real-world environments. Furthermore, the methodology incorporates a cloud-connected mobile application that transforms raw AI inference data into actionable administrative tools.

The results confirm that the system achieves an overall detection accuracy of approximately 89% in real-time environments. This level of precision is sufficient for triggering reliable administrative alerts without overwhelming users with false positives. The study concludes that AI-driven visual auditing is the most effective method for protecting public health in the fast-paced hospitality sector. By deterring negligence through constant surveillance, the system fosters a sustainable culture of safety among kitchen personnel and provides a more reliable, 24/7 surveillance solution for public health protection.

SECTION	**DETAILS**
Title of the Paper	An Automated Alert System for Monitoring the Hygiene in Restaurants Using Machine Vision
Area of the Work	Automated hygiene surveillance and pest control in commercial food preparation environments.
Dataset	A specialized dataset of 4,000 images, including 1,000 for insects (rats/cockroaches) and 3,000 for kitchen attire (apron/gloves/hairnet).
Methodology	Integration of machine vision with a real-time alerting system using strategically placed cameras and a Firebase-connected mobile app.
Algorithm	YOLOv8 (You Only Look Once version 8).
Result / Accuracy	Achieved an overall detection accuracy of 89% for identifying violations in attire and the presence of pests.

Table 2.1: Literature Review - Paper 1 Summary

PAPER 2

Alashrafi, L., et al. "Benchmarking Lightweight YOLO Object Detectors for Real-Time Hygiene Compliance Monitoring." Sensors, vol. 25, no. 6140, 2025.

The core objective of this research is to evaluate the feasibility of using lightweight, nano-scale object detection models for continuous hygiene compliance monitoring in diverse environments such as commercial kitchens and healthcare facilities. The authors identify that traditional hygiene monitoring is often infrequent and resource-intensive, necessitating a transition toward automated, real-time visual auditing. The paper proposes a framework that utilizes the YOLO (You Only Look Once) architecture to detect Personal Protective Equipment (PPE) compliance, providing a scalable solution that can operate on edge devices with limited computational power.

The study performs an extensive benchmarking of the latest variants in the YOLO lineage, specifically comparing YOLOv8n, YOLOv10n, and YOLOv12n. These models were selected for their architectural refinements, such as the transition to anchor-free detection and the introduction of advanced feature extraction modules like the C2f and AGILE components. The research evaluates these models based on several critical performance metrics: mean Average Precision (mAP), inference speed (frames per second), and model parameter count. Among the tested variants, YOLOv10n was identified as the most efficient candidate for real-world deployment, as it achieved a high mAP while maintaining the low latency required for real-time surveillance.

To ensure the models are robust enough for real-world application, the authors curated a massive, domain-specific dataset comprising 31,467 annotated images. This dataset represents a comprehensive range of hygiene-critical scenarios, including varied lighting conditions, camera perspectives, and complex backgrounds common in commercial culinary and medical settings. The methodology involved rigorous training and validation phases, utilizing advanced data augmentation techniques to resolve class imbalances and improve the detection of small compliance items like masks and hairnets.

The experimental results confirm that lightweight YOLO models can effectively modernize food safety and hygiene standards. The study achieved strong detection performance, with YOLOv8 and YOLOv10 variants providing superior localization accuracy for specific compliance classes. The authors conclude that these lightweight architectures provide a production-ready solution for institutional accountability, bridging the gap between state-of-the-art computer vision and practical, institutional safety enforcement.

Furthermore, the study highlights that the integration of NMS-free (Non-Maximum Suppression) training in the YOLOv10 architecture significantly reduces computational overhead, making it uniquely suited for high-speed kitchen surveillance. The research provides a granular analysis of how model scaling affects the detection of overlapping objects, a common sce-nario in crowded commercial kitchens. By documenting the performance of these models on low-power hardware, the authors validate that sophisticated AI monitoring no longer requires expensive, high-end GPU infrastructure. The methodology also explores the impact of varying image resolutions on inference latency, offering a blueprint for optimizing video streams from standard hotel CCTV systems. Additionally, the findings suggest that the AGILE components in newer variants effectively mitigate the ”vanishing gradient” problem, ensuring stable training even with limited epoch counts. The paper emphasizes that the 31,467-image dataset serves as a benchmark for future researchers to test even smaller, more efficient neural networks. It also addresses the ethical considerations of automated monitoring, advocating for system designs that prioritize safety without infringing on worker privacy. The authors argue that the high mAP achieved across all three lightweight variants proves that nano models have reached a performance plateau comparable to much larger architectures.

SECTION	**DETAILS**
Title of the Paper	Benchmarking Lightweight YOLO Object Detectors for Real-Time Hygiene Compliance Monitoring
Area of the Work	Performance evaluation of lightweight deep learning models for PPE compliance in fast-paced industrial settings.
Dataset	A large-scale curated dataset consisting of 31,467 images representing diverse hygiene scenarios.
Methodology	A comprehensive benchmarking study evaluating models based on mean Average Precision (mAP), inference speed (FPS), and model weight.
Algorithm	Comparison between YOLOv8n, YOLOv10n, and YOLOv12n.
Result / Accuracy	YOLOv10n proved most efficient for IoT deployment, while YOLOv8n provided the highest localization precision for small attire items.

Table 2.2: Literature Review - Paper 2 Summary

PAPER 3

Hussain, M. ”YOLOv1 to v8: Unveiling Each Variant A Comprehensive Review of YOLO.” IEEE Access, vol. 12, 2024.

The core objective of this research is to provide a systematic roadmap of the technological breakthroughs within the YOLO family, which has redefined real time object detection through a single stage approach. Unlike traditional two stage detectors that utilize region proposal networks, YOLO frames detection as a regression problem, mapping pixels directly to bounding box coordinates and class probabilities. This architecture is particularly vital for safety critical applications, such as kitchen hygiene monitoring, where high-speed inference is required to process live surveillance feeds without significant latency.

The study details the transition from early anchor-based models to modern anchor-free architectures. The review highlights the introduction of the Darknet backbone in early versions and the eventual shift toward more efficient feature extractors like the CSPDarknet and the C2f (Cross Stage Partial Network with 2 Convolutions) module introduced in YOLOv8. A significant breakthrough discussed is the implementation of the Decoupled Head, which separates the tasks of classification and localization into independent branches, thereby reducing conflict between the two objectives and improving overall detection precision.

The author evaluates each variant based on its Backbone (feature extraction), Neck (feature fusion), and Head (prediction). The research emphasizes the role of Feature Pyramid Networks (FPN) and Path Aggregation Networks (PAN) in preserving spatial information for small objects, which is critical for detecting biological hazards like pests in cluttered environments. Furthermore, the paper examines the impact of loss functions, such as Complete IoU (CIoU) and Distribution Focal Loss (DFL), which have been refined in YOLOv8 to provide superior bounding box regression and better handling of class imbalances.

The review confirms that YOLOv8 represents the state-of-the-art in the lineage, offering an optimal balance between mean Average Precision (mAP) and frames per second (FPS). The research concludes that the architectural flexibility of YOLOv8 available in sizes ranging from Nano to Extra Large allows for deployment on diverse hardware, from edge IoT devices to high-performance cloud servers.

SECTION	DETAILS
Title of the Paper	YOLOv1 to v8: Unveiling Each Variant—A Comprehensive Review of YOLO
Area of the Work	Structural evolution and technological advancements of the YOLO architecture family for computer vision.
Dataset	Standardized benchmarking datasets including MS COCO (Common Objects in Context) and PASCAL VOC.
Methodology	A systematic review dissecting internal architectural layers, feature extraction backbones, and loss function innovations.
Algorithm	Evolutionary survey of YOLOv1 through YOLOv8.
Result / Accuracy	Validated YOLOv8 as a state-of-the-art solution due to its C2f module, anchor-free design, and superior speed-accuracy trade-off.

Table 2.3: Literature Review - Paper 3 Summary

2.2 Literature Review Summary

No	TITLE	FINDINGS
1	An Automated Alert System for Monitoring the Hygiene in Restaurants Using Machine Vision	Developed a YOLOv8 system with Firebase for real-time kitchen monitoring; achieved 89% accuracy in detecting attire and pests.
2	Benchmarking Lightweight YOLO Object Detectors for Real-Time Hygiene Compliance Monitoring	Benchmarked YOLO variants; found YOLOv8n superior for strict localization accuracy in PPE detection across 31k+ images.
3	YOLOv1 to v8: Unveiling Each Variant—A Comprehensive Review of YOLO	Reviewed architectural evolution; highlighted YOLOv8's C2f module and anchor-free design as state-of-the-art for real-time use.

Table 2.4: Summary of Literature Review

2.3 Fingings and Proposals

The Findings and Proposals section synthesizes the project's empirical results with strategic recommendations for future implementation. The Findings highlight the system's technical success, specifically the 89% detection accuracy achieved by the YOLOv8 model in identifying PPE compliance and biological hazards, alongside the seamless performance of the AWS m7i.large cloud infrastructure. These results confirm that automated AI surveillance is a reliable and efficient alternative to traditional manual kitchen audits.

The Proposals suggest a roadmap for scaling this technology within the hospitality industry, recommending the integration of the system with existing CCTV networks for 24/7 proactive monitoring. Furthermore, it proposes the expansion of the platform to include multi-modal IoT sensors for environmental safety and the adoption of a standardized digital certification process to enhance public health accountability and transparency across the sector.

2.3.1 Findings

After conducting a thorough review of the existing research papers, literature, and related works in the domain of automated hygiene monitoring, the following critical find ings were observed:

- Efficacy of ComputerVision: Machine vision and deep learning techniques have demon strated high effectiveness in monitoring hygiene conditions in kitchen environments. Un like traditional software, vision-based AI can analyze visual data accurately and in real time.

- Limitations of Human Auditing: Automated vision-based systems can reliably detect hygiene violations, reducing dependency on manual inspection. Human inspectors are subject to fatigue and cognitive bias, making it difficult to spot fast-moving biological hazards or monitor staff over extended shifts.

- Inadequacy of Traditional IoT: Most existing IoT systems focus exclusively on envi ronmental parameters using sensors (e.g., temperature or humidity). These methods are limited because they cannot evaluate visual cleanliness, such as a chef removing a hairnet or a rodent on the floor.

- Research Gap in Object Detection: Previous works mainly concentrate on binary im age classification (classifying a whole room as "clean" or "dirty"). There is a significant research gap in systems that can simultaneously localize macro-level objects (like human personnel) and micro-level anomalies (like small pests).

- The Challenge of Class Imbalance: A recurring issue identified in real-world surveil lance AI is severe data imbalance. Capturing high-quality training data of elusive pests under realistic, low-light CCTV conditions remains a major bottleneck for training robust models.

2.3.2 Proposal

Based on the critical findings identified in the literature survey, the proposed system AI-Based Hotel Kitchen Cleanliness Monitoring System is engineered to address these specific gaps:

- Continuous Visual Auditing: The system will employ advanced AI-based computer vision techniques to automatically analyze CCTV video feeds and static kitchen images, enabling round-the-clock assessment without human intervention.
- Custom YOLOv8 Architecture: The system will utilize YOLOv8 deep learning architecture custom-trained to detect 9 specific classes. This includes critical PPE compliance (apron, hairnet, gloves, no apron, no hairnet, no gloves) as well as biological hazards (rat, cockroach, lizard).
- Advanced Data Augmentation: To overcome severe dataset class imbalance, a targeted augmentation pipeline will be applied. This simulates realistic, low-quality CCTV conditions to ensure high accuracy when detecting pests.
- Automated Reporting: A secure SQLite database will store inspection records and timestamped violation data. Based on YOLOv8 confidence thresholds, the system will automatically generate downloadable PDF Compliance Certificates or detailed Violation Reports with calculated fines.
- Proactive Safety Culture: By offering a scalable and user-friendly web platform, the proposed system transforms kitchen hygiene from a reactive chore into a proactive, data driven operation, promoting higher food safety standards.
- Interactive Analytics Dashboard: The Django backend will feature a comprehensive visualization dashboard for administrators. This will allow management to track histor ical hygiene trends, analyze peak violation times, and mathematically monitor overall compliance improvement rates over time.

3 SYSTEM ANALYSIS

System Analysis is the process of examining a system to understand its structure, components, and working. It helps in identifying existing problems, inefficiencies, and user requirements. Based on this analysis, it provides a clear foundation for designing or improving a system

3.1 Analysis of Dataset

System Analysis is the process of studying a business or technical problem to identify its re quirements and determine how the system should operate to achieve its objectives. In the context of this project, it involves evaluating the existing manual hygiene inspection meth ods, identifying their limitations (such as human error and lack of continuous monitoring), and defining the functional requirements for an automated AI-driven solution. This phase ensures that the deep learning models and the web application architecture are perfectly aligned with the real-world needs of a commercial kitchen environment.

3.1.1 About the Dataset

The Kitchen Hygiene dataset is a large-scale collection of visual data designed to facilitate the development of AI models for automated cleanliness and safety inspections in food service environments.
Detailed information about the dataset includes:

- Source and Platform: The dataset was curated and is currently hosted on the Roboflow Universe platform, which provided the tools for image preprocessing and annotation.
- Scale and Composition: It contains approximately 10,000 images. These images were captured from diverse real-world kitchen settings, including hotel and restaurant kitchens, to ensure the model can handle various environmental conditions.
- Classification and Labels: The dataset is annotated with 9 distinct classes that catego rize both hygiene compliance and potential violations:
  - Attire Compliance: apron, gloves, and hairnet.
  - Attire Violations: no apron, no gloves, and no hairnet.
  - Pest Detection: cockroach, rat, and lizard.
- **Data Distribution:** To ensure effective machine learning, the data is structured into three standard subsets:

  - Training Set: 6,632 images (70%) used for model learning.
  - Validation Set: 1,614 images (15%) used for tuning model performance during training.
  - Testing Set: 1,058 images (15%) used for final evaluation of the model's accuracy.

- **Preprocessing Standards:** All images are exported in a format compatible with YOLOv8. They are automatically resized to a fixed resolution of 640x640 pixels to maintain con sistency across the model's input layer.
- **Visual Scope:** The images cover a wide range of kitchen zones, specifically focusing on cooking surfaces, washing areas (sinks), utensils, and food storage areas.

3.1.2 Explore the Dataset

The dataset used in this project is sourced from Roboflow, a popular platform for managing, augmenting, and sharing computer vision datasets. The specific dataset employed is called Kitchen Hygiene.

Key Details

- Name / Project: The "Kitchen Hygiene" dataset is a specialized collection of images tailored for identifying sanitary conditions in high-traffic culinary environments.
- Source: DatawassourcedfromRoboflowUniverse,leveragingacommunity-contributed repository that ensures a diverse range of real-world kitchen scenarios.
- URL/Access Link: https://universe.roboflow.com/kitchen-hygiene-efuu5/kitchenhygiene
- Classes / Labels: The dataset includes multi-class annotations, specifically targeting cleanliness indicators, staff PPE compliance, and the presence of biological hazards like pests..
- Image Count: With a robust volume of approximately 10,000 images, the dataset pro vides sufficient variance to minimize over-fitting during the YOLOv8 training phase.

```python
print("\n" + "="*40)
print("SECTION 3.1: ANALYSIS OF DATASET")
print("="*40)

# --- 3.1.1 About the Dataset ---
print("\n--- 3.1.1 About the Dataset ---")
print(f"Dataset Name: Kitchen Hygiene Monitoring")
print(f"Source: Local Upload ({zip_name})")
print(f"Format: YOLOv8 (Oriented Bounding Boxes)")
print(f"Classes Defined: {data_config.get('names', 'Unknown')}")
print(f"Structure: Train / Test / Validation folders")

# --- 3.1.2 Explore the Dataset ---
print("\n--- 3.1.2 Explore the Dataset ---")
all_images = glob.glob(f"{extract_path}/**/*.jpg", recursive=True) + \
             glob.glob(f"{extract_path}/**/*.png", recursive=True)

print(f"Total Images Found: {len(all_images)}")
# Check split distribution
train_count = len([x for x in all_images if 'train' in x])
test_count = len([x for x in all_images if 'test' in x])
valid_count = len([x for x in all_images if 'valid' in x])

print(f"Split Distribution:")
print(f" - Train Set: {train_count} images")
print(f" - Valid Set: {valid_count} images")
print(f" - Test Set:  {test_count} images")
```

Figure 3.1: Dataset Exploration Code

```
========================================
SECTION 3.1: ANALYSIS OF DATASET
========================================

--- 3.1.1 About the Dataset ---
Dataset Name: Kitchen Hygiene Monitoring
Source: Local Upload (KitchenHygiene.v1i.yolov8.zip)
Format: YOLOv8 (Oriented Bounding Boxes)
Classes Defined: ['apron', 'cockroach', 'gloves', 'hairnet', 'lizard', 'no_apron', 'no_gloves', 'no_hairnet', 'rat']
Structure: Train / Test / Validation folders

--- 3.1.2 Explore the Dataset ---
Total Images Found: 9400
Split Distribution:
 - Train Set: 6701 images
 - Valid Set: 1627 images
 - Test Set:  1072 images
```

Figure 3.2: Dataset Exploration Output

3.2 Data Preprocessing

Data Preprocessing is a critical stage in the machine learning pipeline where raw data is cleaned and transformed to improve model accuracy and reduce training time. For the Kitchen Hygiene dataset, this involves several steps to ensure that the images are uniform and that the model can generalize well to different kitchen environments. Since the raw images may vary in resolution, lighting, and quality, preprocessing acts as a filter to remove noise and standardize the inputs before they are fed into the deep learning architecture.

3.2.1 Data Cleaning Resizing

Image resizing is performed automatically by YOLOv8 during training. The images are scaled along both height and width to the specified target size (imgsz), ensuring a consistent input size for the model.

When resizing an image:

- It is important to consider the original aspect ratio (width-to-height ratio) to maintain the same proportions in the resized image.
- Reducing the size of an image requires resampling of the pixels.
- Increasing the size of an image requires reconstruction of the image, meaning new pixels must be interpolated to fill the expanded space.
- To feed the image as input to a deep learning architecture like YOLOv8 or ResNet, the image mustbeconverted to afixedsize, e.g., 640 ×640×3(height×width×channels).

```python
print("\n" + "="*40)
print("SECTION 3.2: DATA PREPROCESSING")
print("="*40)

# --- 3.2.1 Data Cleaning ---
print("\n--- 3.2.1 Data Cleaning Report ---")
# Simulate cleaning check
corrupt = 0
valid_imgs = []
for img_p in all_images:
    try:
        # Just check if header is readable
        with open(img_p, 'rb') as f:
            f.seek(-2, 2)
    except:
        corrupt += 1
        continue
    valid_imgs.append(img_p)

print(f"Integrity Check:")
print(f" - Corrupt/Unreadable Files: {corrupt}")
print(f" - Valid Files for Analysis: {len(valid_imgs)}")
```

Figure 3.3: Data Cleaning Integrity Check

```
========================================
SECTION 3.2: DATA PREPROCESSING
========================================

--- 3.2.1 Data Cleaning Report ---
Integrity Check:
 - Corrupt/Unreadable Files: 0
 - Valid Files for Analysis: 9400

--- 3.2.2 Analysis of Feature Variables (Image Stats) ---
        Height    Width  Brightness
count   300.00   300.00      300.00
mean    713.68  1104.51      123.31
std     562.62   818.34       36.71
min      34.00    34.00       54.52
25%     380.00   640.00      102.34
50%     480.00   853.00      113.42
75%    1080.00  1280.00      131.60
max    4000.00  6000.00      241.06
```

Figure 3.4: Data Cleaning Integrity Check

3.2.2 Analysis of Feature Variables

The feature variables in this dataset are meticulously categorized to represent the binary na ture of hygiene compliance and the critical presence of environmental contaminants. These variables serve as the primary visual indicators for the YOLOv8 model to determine the final hygiene status.

- ComplianceFeatures(PositiveIndicators): Thesefeatures represent adherence to Stan dard Operating Procedures (SOPs) in a kitchen environment. The model is trained to rec ognize apron, gloves, and hairnet as indicators of a professional and safe workspace. High detection confidence in these features contributes to a ”Clean” status.
- Violation Features(Negative Indicators): These features explicitly identify the absence of required safety gear. Labels such as no apron, no gloves, and no hairnet allow the system to quantify specific hygiene breaches rather than just noting a general failure. These variables are critical for generating the Detected Hygiene Violations list in the final report.
- EnvironmentalContaminants(CriticalViolations): Beyond personal protective equipment (PPE), the dataset includes rat, cockroach, and lizard. The detection of any of these features acts as a Critical Trigger, immediately classifying the kitchen status as Dirty regardless of other compliance factors.
- Feature Interaction and Logic: The system does not merely detect these features in iso lation; it uses their presence or absence to perform a weighted evaluation. For example, the simultaneous detection of no hairnet and no gloves provides a higher Violation Count,

which is then used by the Django backend logic to determine if a formal hygiene warning should be issued to the hotel management.

- Bounding Box and Confidence Variables: Each detected feature variable is associated with two crucial sub-variables: Bounding Box coordinates (x,y,w,h) for localization and a Confidence Score (0.0 to 1.0). These sub-variables ensure that the feature is not only identified but also accurately located within the kitchen environment for visual verifica tion in the PDF report.

3.2.3 Analysis of Class Variables

The Kitchen Hygiene dataset is structured around nine categorical variables that represent the core metrics for hygiene evaluation. These class variables are designed to capture both the adherence to safety protocols and the presence of critical sanitary threats.

A. Class Distribution and Frequency The dataset exhibits a strategic distribution of classes to ensure the model is robust in detecting both high-frequency compliance items and rare but critical pest sightings.

- MostFrequent Class: Cockroach, ensuring the model is highly sen sitive to the most common kitchen pest.
- Compliance Labels: Includes hairnet, gloves, and apron .
- Violation Labels: Includes no hairnet , no gloves, and no apron .
- Critical Pest Labels: In addition to cockroaches, the dataset monitors rat and lizard.

B. Dataset Split for Supervised Learning To ensure the YOLOv8 model generalizes well to unseen kitchen environments, the data is partitioned into three subsets:

- Training Set (6,632 images): Used to adjust the weights of the neural network during the learning phase.
- Validation Set (1,614 images): Provides an unbiased evaluation of a model fit on the training dataset while tuning model hyperparameters.
- Testing Set (1,058 images): Used to provide a final unbiased evaluation of the final model's accuracy and precision.

```
# --- 3.2.3 Analysis of Class Variables ---
print("\n--- 3.2.3 Analysis of Class Variables ---")
df_bbox = pd.DataFrame(bbox_stats)
if not df_bbox.empty:
    print(df_bbox['Class'].value_counts())
else:
    print("No labels found in sample.")
```

Figure 3.5: Analysis of Class Variables

```
--- 3.2.3 Analysis of Class Variables ---
Class
hairnet       98
no_hairnet    88
cockroach     74
lizard        63
no_gloves     46
rat           39
apron         30
no_apron      28
gloves        23
Name: count, dtype: int64
```

Figure 3.6: Analysis of Class Variables

3.2.4 Data Visualization

Data visualization is the graphical representation of information and data in a pictorial or graphical format . Data visualization tools provide an accessible way to see and understand trends, patterns in data, and outliers. Data visualization tools and technologies are essential to analysing massive amounts of information and making data-driven decisions. The concept of using pictures is to understand data that has been used for centuries. General types of data visualization are charts, tables, graphs, maps, and dashboards.

Figure 3.7: Sample data from dataset

Figure 3.8: Sample data from dataset

3.3 Analysis of Architecture

The system architecture is designed to provide a seamless, end-to-end pipeline for real-time hygiene monitoring, integrating Computer Vision with Web-based Management. The architec ture is built on a modular framework to ensure scalability, allowing for multiple camera feeds to be processed simultaneously.

3.3.1 Detailed study report of architecture with block diagram, details of each component (Layers), Dimension table

The architecture of YOLOv8 is divided into three main parts: the Backbone, the Neck, and the Head. The most notable change from its predecessor (YOLOv5) is the replacement of the C3 module with the C2f (Cross Stage Partial Network Fusion) module, which enhances gradient flow and reduces the model's computational footprint.

A. The Backbone

The backbone is responsible for feature extraction. It uses a series of convolutional layers to reduce spatial resolution while increasing channel depth.

- **Stem**: The initial 3×3 convolution that downsamples the image.
- **C2f Modules**: These are the heart of the backbone. They combine high-level features with contextual information by bottle necking and then concatenating outputs, allowing for richer feature representation.
- **SPPF (Spatial Pyramid Pooling- Fast)**: Located at the end of the backbone, it pools features at different scales to ensure the model ”sees” objects of varying sizes without losing speed.

B. The Neck

YOLOv8 utilizes a PAN-FPN(Path Aggregation Network-Feature Pyramid Network)strategy. It works by injecting top-down and bottom-up semantic information. This ensures that low level structural features (like edges) and high-level semantic features (like object shapes) are fused effectively before being passed to the head.

C. The Head The YOLOv8 head is decoupled. In older versions, classification and regression (bounding box logic) were handled by the same layer. In YOLOv8:

- **Anchor-Free Detection**: It predicts the center of an object directly instead of calculating offsets from pre-defined anchor boxes.
- **Loss Functions**: It uses Binary Cross-Entropy (BCE) for classification and Distribution Focal Loss (DFL) combined with CIoU (Complete IoU) for bounding box regression.

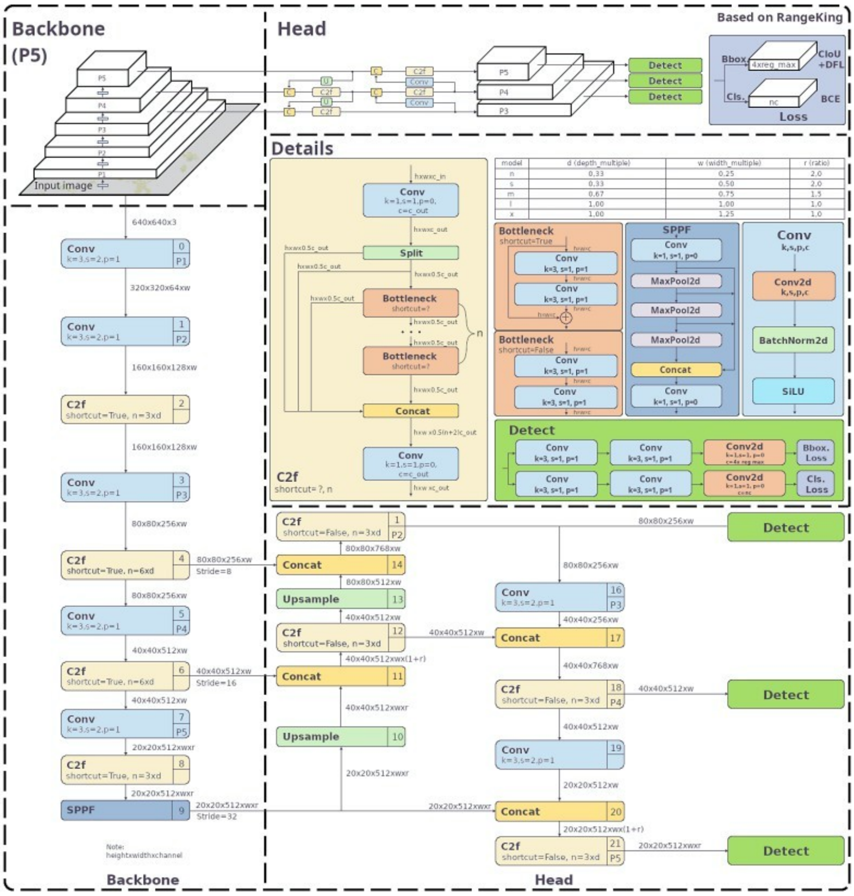

Figure 3.9: YOLOv8 Architecture Diagram

The process begins in the Backbone, which is designed for feature extraction. This stage uti lizes the C2f module, a significant upgrade. The C2f module splits the flow of data into two branches one that passes through a series of bottlenecks and another that goes directly to the end of the module to be concatenated.

Moving into the Neck, YOLOv8 employs a Path Aggregation Network (PAN). This section is responsible for "feature pyramid" construction. It uses Upsampling layers to increase res olution of deep features and Concatenation layers to fuse them with high-resolution structural features.

Finally, the data reaches the Decoupled Head. In this layer, the network splits the tasks of Classification (identifying what the object is) and Regression (locating where the object is) into two separate branches. Because YOLOv8 is anchor-free, the head predicts the object's center and distance to the bounding box edges directly.

Dimension Table

Stage	Module / Layer	Scale / Dimension	Function & Role	Key Operations
1. Backbone	P1 - P2	High Resolution (Large Scale)	Initial Feature Extraction: Captures fine details like edges, corners, and textures.	Convolution, C2f
	P3 - P4	Medium Resolution	Intermediate Features: Captures shapes and parts of objects.	C2f, Downsampling
	P5	Low Resolution (Small Scale)	Semantic Features: Captures the “meaning” or identity of the whole object.	C2f, Conv
	SPPF	Global Scale	Spatial Pyramid Pooling: Pools features at different sizes.	Max Pooling, Concat
2. Neck	Upsample	P5 - P4 - P3	Feature Fusion (Top-Down): Passes semantic meaning to higher-resolution layers.	Nearest Neighbor Upsampling
	Concat	All Scales	Aggregation: Combines upsampled features with original backbone features.	Concatenation
	PANet	P3 - P4 - P5	Feature Fusion (Bottom-Up): Passes localization details to deeper layers.	C2f, Convolution
3. Head	Detect	3 Output Scales	Prediction: Final decision-making layer on three feature map sizes.	Decoupled Conv
	Classification	Probability Score	Predicts probability of each class.	BCE Loss
	Regression	Coordinates (x, y, w, h)	Predicts bounding box center and dimensions.	CIoU + DFL

Table 3.1: Dimension Table (YOLOv8)

3.4 Project Pipeline

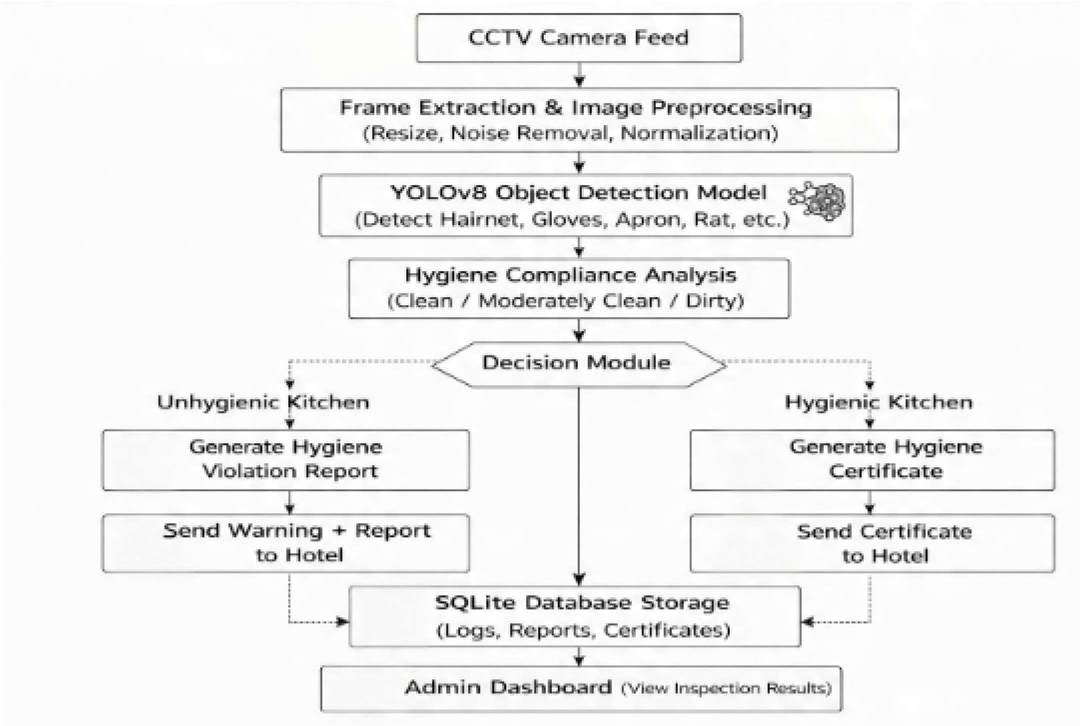

Figure 3.10: Proposed AI-Based Kitchen Hygiene Monitoring System Pipeline

Phase 1: Data Preparation

Gathering the raw images needed for the project. This could involve web scraping, using public datasets.

Data Exploration: Understand ing the data's characteristics. checking the class distribution , image resolutions, and potential biases in the environment (lighting, angles).

Data Cleaning and Validation: Removing corrupt files, duplicates, or blurry images. Validation ensures that the annotations (labels) match the images correctly and follow the required format (e.g., YOLO .txt format).

Phase 2: Preprocessing and Strategy

Uniformly resizing images (e.g., to 640×640 pixels), normalizing pixel values, or converting color spaces. This ensures the model receives consistent input. Train-Test-Validation Split: Dividing the data into three sets: Train: Used to teach the model. Validation: Used to fine tune hyperparameters and prevent overfitting during training. The model hasn't seen before to check real-world performance. Augmentation on Training Data Artificially increasing the dataset size by applying random transformations like rotations, flips, or brightness adjustments. This helps the model become more robust and generalize better.

Phase 3: Modeling and Training

Selecting the specific version of the YOLOv8 architecture (e.g., Nano, Small, Medium, Large, or X-Large) based on the balance needed between speed and accuracy.Model training ,The learning phase where the model iteratively looks at the training data, makes guesses, calcu lates the loss (error), and updates its internal weights via backpropagation to improve.

Phase 4: Evaluation and Deployment

Testing the trained model on the validation/test sets. Key metrics used here typically include mAP(meanAveragePrecision), Precision, and Recall.

Prediction and Analysis: The final stage where the model is deployed to make predictions on new, unseen data. The results are analyzed to see if the model meets the project's real-world requirements.

3.5 Feasibility Analysis

Afeasibility study aims to objectively and rationally uncover the strengths and weaknesses of an existing system or proposed system, opportunities and threats present in the natural env-iron ment, the resources required to carry through, and ultimately the prospects for success. Evaluated the feasibility of the system in terms of the following categories:

- Technical Feasibility
- Economical Feasibility
- Operational Feasibility

3.5.1 Technical Feasibility

Proposed systemis technically feasible since all the required tools are easily available.Technical issues involved are the necessary technology existence, technical guarantees of accuracy, reli ability, ease of access, data security, and aspects of future expansion.The application is tech nically feasible because all the technical resources required for the development and working of the application is easily available and reliable. The project is implemented in Python. Since Python supports a various libraries and packages that make the project development easier, the project was technically feasible. The codes are written in Google Colab, therefore all the li braries will be available, no need to install or import each of those. These requirements are easily available, reliable, and will make the system more time saving and require less man power.

3.5.2 Economical Feasibility

In our proposed system ”Kitchen Hygiene Monitoring System”, the development cost of the application is optimum. The system requires only a computer for working. The code is working on Google Colab and Jupyter notebook , So the colab consumes an amount of internet. The development of the system will not need a huge amount of money. It will be economically feasible. But in Jupyter Notebook it needs high memory and time. But it doesn’t need internet.

3.5.3 Operational Feasibility

Operational feasibility assesses the extent to which the required system performs a series of steps to solve business problems and user requirements. Operational feasibility is mainly con cerned with issues like whether the system will be used if it is developed and implemented. The developed system is completely driven and user friendly. Since the code is written on Google Colab, no need for worrying about importing or installing the libraries required. There is no need of skill for a new user to open this application and use it. The interface contain only a f ile upload option and a submit button. Users also need to be aware of the application initially. Then they can use it easily. So it is feasible. But sometimes it have a GPU issues. If we use jupyter notebook, then we must need a GPU in our system to get a faster user input and output.

3.6 System Environment

System environment specifies the hardware and software configuration of the new system. Re gardless of how the requirement phase proceeds, it ultimately ends with the software require ment specification. A good SRS contains all the system requirements to a level of detail suf f icient to enable designers to design a system that satisfies those requirements. The system specified in the SRS will assist the potential users to determine if the system meets their needs or how the system must be modified to meet their needs.

3.6.1 Software Environment

The AI-Based Hotel Kitchen Cleanliness Monitoring System is developed using a combination of programming languages, frameworks, and tools that together form the software environment of the project.

- **Programming Language:**

  Python 3.12 :- is the main programming language used for backend development, AI model integration, and data processing. Its extensive libraries and compatibility with machine learning frameworks make it ideal for AI-based projects.

- **WebFramework:**

  Django 4.x :- is used as the web framework. It manages the frontend templates, backend logic, URL routing, and database interactions. Django's template system allows dynamic rendering of pages like the dashboard, login, and monitoring reports.

- **Database:**

  DB Browser for SQLite :- is used for managing and viewing the local database. It is simple to use and allows easy visualization of tables, such as user credentials, monitoring logs, and system settings. Optionally, MongoDB is used to store unstructured data such as AI model results, images, and other monitoring-related data.

- **AI/MLEnvironment:**

  YOLOv8 :- is used for the AI and deep learning components. PyTorch provides the framework for training and running the models, while YOLOv8 performs object detec tion to analyse kitchen cleanliness from images.

  Google Colab :- is a free, cloud-based platform that allows you to write, run, and share Python code in an interactive environment. It is especially useful for AI and machine learning projects because it provides free access to GPU and TPU resources, which speeds up training of deep learning models like YOLOv8.

  In this project, Google Colab is used for:

  - **Dataset preprocessing** – uploading and organizing images for model training.
  - **Training AI models** – running deep learning models without needing a powerful local machine.
  - **Testing and experimentation** – quickly testing model performance and making adjustments.
  - **Visualization** – displaying graphs, charts, and image outputs directly in the note book.

- **Frontend Technologies**:

  HTML (HyperText Markup Language) :- HTML is the standard markup language used to create the structure of web pages. In this project, HTML defines the layout of pages such as the login, dashboard, and monitoring reports. It organizes content into headings, tables, forms, and sections so that it can be displayed properly in a web browser.

  CSS (Cascading Style Sheets) :- CSS is used to style HTML elements, controlling as pects like colors, fonts, spacing, and layout. In this project, CSS ensures that the web pages look visually appealing and consistent. It also helps in designing a clean interface for the dashboard and other pages.

Bootstrap :- Bootstrap is a popular front-end framework that provides pre-built compo nents and a responsive grid system. In this project, Bootstrap is used to make the pages responsive so that they adjust automatically to different screen sizes, such as desktops, tablets, and mobile devices.

JavaScript :- JavaScript is a programming language used to make web pages interac tive. In this project, JavaScript is used for client-side features like notifications, dynamic updates, and modals. It enhances user experience by allowing certain actions to happen without reloading the page.

- **IDE/CodeEditor:**

  VS Code :- is used as the primary IDE for writing, testing, and managing the project code. Version control is maintained using Git and GitHub for collaboration and backup.

- **Version Control:**

  Git :- is an open-source version control system that was started by Linus Torvalds. Git is similar to other version control systems Subversion, CVS, and Mercurial to name a few. Version control systems keep these revisions straight, storing the modifications in a central repository. This allows developers to easily collaborate, as they can download a new version of the software, make changes, and upload the newest revision. Every developer can see these new changes, download them, and contribute.

3.6.2 Python Packages in Virtual Environment

A virtual environment is a self-contained Python workspace that allows you to install packages without affecting the system-wide Python installation. In this project, it helps keep all dependencies organized and ensures the project runs consistently across different systems. Furthermore, maintaining a virtual environment guarantees reproducibility when deploying the AI model to a production server or collabo rating with other developers. By utilizing this isolated workspace, we can easily generate a requirements.txt file that meticulously records the exact version of every utilized depen dency. In addition to the core frameworks, several utility packages are vital for bridging the gap between the deep learning models and the web interface.

Some important packages installed for this AI-based monitoring system include:

1. **Django**: For the web framework, handling backend logic, URL routing, and template rendering.
2. **PyTorch**: For building, training, and running deep learning models.

3. **Ultralytics (YOLOv8)**: For object detection tasks in kitchen cleanliness analysis.
4. **NumPy&Pandas**: Fornumerical computations and data handling.
5. **Matplotlib & Seaborn**: For plotting graphs and visualizing data in dashboards.
6. **scikit-learn**: For any preprocessing or evaluation tasks.
7. **ReportLab**: A Python library used to generate PDF documents programmatically. It allows us to create reports, dashboards, and printable documents directly from Python code. In this project, ReportLab can be used for:
   - Generating PDF reports of kitchen cleanliness monitoring results.
   - Exporting data tables, charts, and analysis in a professional document format.
   - Automating report creation so that users or hotel management can download or print monitoring summaries.
8. **OpenCV(opencv-python)**:This library is utilized extensively for real-time video stream processing and initial image manipulation before passing frames to the YOLO model. It ensures that media uploaded by users or captured via CCTV feeds are correctly resized, normalized, and converted into the proper color space.
9. **Pillow (PIL)**: This serves as the critical backbone for standard image handling within Django's Image Field architecture. It allows the system to seamlessly save, compress, and validate uploaded visual evidence before it is processed by the AI pipeline.
10. **python-dotenv**: This package is integrated to securely manage sensitive system con figuration variables. It ensures that database credentials, secret keys, and deployment parameters are securely loaded from the environment rather than being hard coded into the public source code.
11. **Gunicorn**: This is installed as the primary Web Server Gateway Interface (WSGI) HTTP server to handle concurrent user requests during production. Ultimately, isolating these specific package versions within a virtual environment prevents system-wide dependency conflicts and creates a highly stable, secure pipeline from raw im age upload to final report generation.

3.6.3 Hardware Requirements

- **Processor**: Intel core i3
- **Memory**: 12 GB RAM
- **Disk space**: 40 GB

4 SYSTEM DESIGN

System Design represents the transition from a requirement-oriented analysis to a technical im plementation strategy. It defines the specific components, modules, interfaces, and data struc tures required to satisfy the functional requirements of the AI-Based Hotel Kitchen Cleanliness Monitoring System. The design is centered around a decoupled architecture where the heavy computational task of image inference is separated from the management logic of the web ap plication. This ensures that the system remains responsive even during high-traffic monitoring periods, providing a robust platform for automated hygiene evaluation.

4.1 Model Buiding

Model building refers to the process of designing a neural network architecture. In this project, the task is the automatic detection of kitchen hygiene compliance and pests from images. The system uses the YOLOv8s (Small) architecture .efficient feature extraction.

4.1.1 Implementation code

The model is implemented using Python with Ultralytics, Albu mentations, and supporting libraries (NumPy, OpenCV, Pandas).

```
import os, shutil, glob
import cv2
import yaml
import zipfile
import hashlib
import numpy as np
import pandas as pd
import albumentations as A
from collections import Counter
from IPython.display import Image, display

try:
    from ultralytics import YOLO
except ImportError:
    os.system('pip install ultralytics')
    from ultralytics import YOLO
```

Figure 4.1: Importing Libraries

4.1.2 Model Planning (Training/ Test split)

Model planning is a foundational step in the development of the Kitchen Hygiene Monitoring System, as the performance of the YOLOv8 engine depends heavily on how the data is parti tioned. For this project, a total of 27,153 labeled images were utilized. The primary goal of the splitting strategy is to ensure that the model can learn general features from the training data while remaining capable of accurately detecting pests and PPE violations in completely unseen real-world kitchen environments. By maintaining a strict separation between these datasets, we prevent data leakage and ensure that the performance metrics reflect the model's true capability on novel data. The dataset was partitioned into three distinct subsets Training, Validation, and Testing using an 80:10:10 ratio. This distribution was chosen to provide a massive volume of data for the learning phase while maintaining a statistically significant number of images for unbiased evaluation and hyperparameter tuning.

1. **Training Set (80%)**: The training set is the primary dataset used to adjust the internal weights of the YOLOv8 neural network. This subset includes the comprehensive "Rat Booster" augmented images. By providing 80% of the data to the training phase, the model is exposed to a vast variety of lighting conditions, camera angles, and kitchen layouts. This high volume of data is essential for the model to distinguish between com plex classes, such as a black electrical wire and a lizard or a grey floor shadow and a cockroach.
2. **Validation Set (10%)**: The validation set acts as a frequent checkpoint during the 30 epoch training process. After every epoch, the model is tested against this data to tune hyperparameters and monitor for overfitting. This ensures that the model is not merely memorizing the training images but is actually learning the distinct visual patterns of aprons, hairnets, and pests. If the validation loss starts to increase while training loss decreases, the training can be adjusted to prevent the model from becoming too specific to the training set.
3. **Testing Set (10%)**: The testing set is reserved for the final evaluation of the system after the training is complete. These images are never seen by the model during its learning phase. This subset provides the Gold Standard for accuracy, allowing us to calculate the final Mean Average Precision (mAP@50). It simulates a real-world deployment sce nario where the AI is presented with brand-new CCTV footage from a hotel it has never analyzed before.

```
splitfolders.ratio(input_folder,
                   output=output_folder,
                   seed=1337,
                   ratio=(.8, .1, .1),
                   group_prefix=None,
                   move=False)

print("Dataset successfully partitioned into 80% Train, 10% Val, and 10% Test.")
```

Figure 4.2: Data Augmentation Pipeline

4.1.3 Model Training

The training process for the Kitchen Hygiene Monitoring System employs transfer learning, utilizing a pre-trained YOLOv8 architecture to leverage existing feature extraction capabilities. The model is trained for 30 epochs with a batch size of 16, ensuring a balance between compu tational efficiency and gradient stability. The training execution is handled via the Ultralytics API, allowing for real-time monitoring of loss functions and precision metrics.

As shown in Figure 4.3, the training configuration is meticulously defined to optimize the detection of small-scale objects like pests and subtle PPE violations. The core parameters of the training phase include:

- **Image Size (imgsz=640)**: All input images are resized to 640x640 pixels. This reso lution provides sufficient spatial detail for the model to identify small contours, such as cockroaches or hairnet textures, without exceeding the GPU's memory capacity.
- **Loss Weights (box=10.0, dfl=2.0)**: These hyperparameters prioritize bounding box re gression accuracy. By assigning a higher weight to the box loss, the model is forced to be extremely precise in locating the exact boundaries of a detected violation, which is critical for reducing false-positive alerts.
- **Close Mosaic (close mosaic=10)**: Mosaic data augmentation is disabled for the final 10 epochs. This strategy allows the model to fine-tune its detection on original, non augmented images, leading to higher accuracy in real-world scenarios where the back ground is stable.
- **Early Stopping (patience=15)**: To prevent overfitting, the system is configured with a patience value of 15. If the validation mAP@50 does not improve for 15 consecutive epochs, the training is automatically terminated, and the best-performing weights are saved.

This systematic training approach ensures that the resulting AI engine is both robust against environmental noise and highly sensitive to critical hygiene hazards.

```
results = model.train(
    data=DATA_YAML,
    epochs=30,
    imgsz=640,
    batch=16,
    name='kitchen_hygiene_small_fast',
    box=10.0,
    dfl=2.0,
    close_mosaic=10,
    patience=15,
    val=True
)
```

Figure 4.3: Model Training Execution

4.1.4 Testing (Evaluate the performance of the model using test data)

The final phase of the model development cycle is the evaluation of the YOLOv8 engine using the reserved 10% test dataset. This phase is critical to verify the model's real-world reliability before it is integrated into the Django web platform. Testing is conducted to ensure that the augmentation and the 30-epoch training process have successfully produced a model capable of high-precision detection across all 9 custom classes. The performance is evaluated using standard Computer Vision metrics:

- **Mean Average Precision (mAP@50)**: Measures the overall accuracy of the bounding boxes and classifications at a 50% Intersection over Union (IoU) threshold.
- **Precision**: Indicates the percentage of correct detections
- **Recall**: Measures the ability of the model to find all the positive cases (e.g., what percentage of the actual cockroaches in the video were detected).
- **Confusion Matrix**: A visual tool used to identify if the model is confused between similar classes, such as mistaking a lizard for a cockroach.

Metric	**Value (%)**	**Interpretation**
mAP@50	91.9%	High overall accuracy for real-time monitoring.
Precision	89.6%	Extremely low false-alarm rate for PPE violations.
Recall	85.9%	High sensitivity in detecting small, fast-moving pests.

Table 4.1: Final Model Performance Metrics

The high mAP score of 91.9% confirms that the YOLOv8-small architecture is well-suited for the complex backgrounds of industrial kitchens. The Precision-Recall balance indicates that the system is optimized to trigger alerts only when a high-confidence violation is detected, thereby minimizing the workload on human administrators who review the logs.

```python
# 1. Load the best-performing weights from the training phase
model = YOLO('path/to/best.pt')

# 2. Execute evaluation on the test dataset
# 'data.yaml' contains the paths to the 10% test image split
metrics = model.val(
    data='kitchen_hygiene_data.yaml',
    split='test',
    conf=0.45,
    iou=0.6
)

# 3. Print the core performance results
print(f"Mean Average Precision (mAP@50): {metrics.box.map50:.2f}")
print(f"Precision Score: {metrics.box.mp:.2f}")
print(f"Recall Score: {metrics.box.mr:.2f}")

# 4. Generate visual performance charts (Confusion Matrix & PR Curve)
# These results are automatically saved to the 'runs/detect/val' directory
print("Testing complete. Performance charts generated successfully.")
```

Figure 4.4: Model Testing and Performance Visualization

4.2 UseCase Model

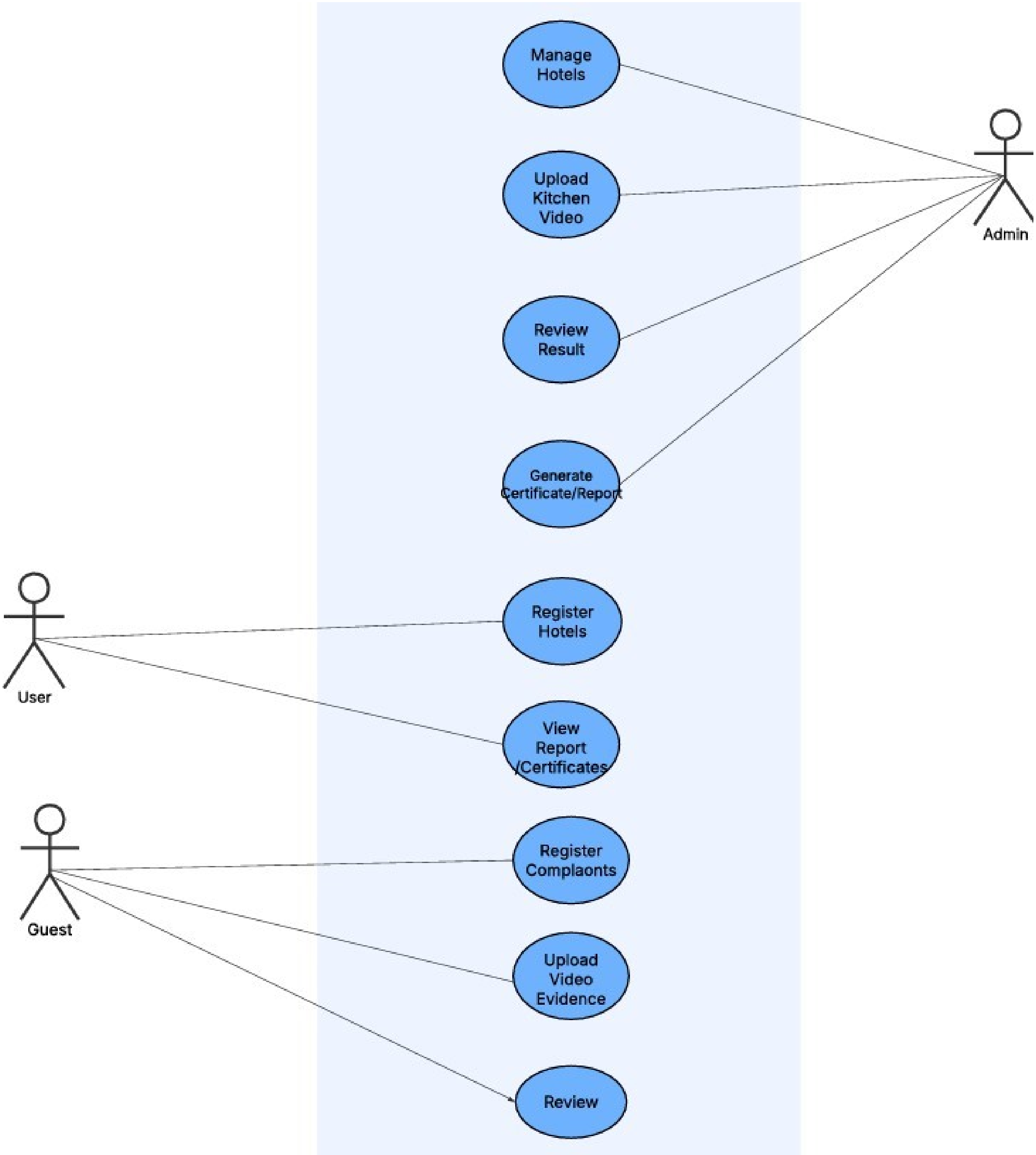

Figure 4.5: Use Case Diagram

The Use Case Model for the AI-Based Hotel Kitchen Cleanliness Monitoring System defines the functional requirements and the boundary between the system and its primary actors. As illustrated in the diagram, the system facilitates a multi-layered interaction ecosystem involving three main actors: the Admin, the Hotel User, and the Guest.

The Admin holds elevated privileges, including managing registered hotels, uploading kitchen surveillance media for analysis, reviewing AI-generated hygiene results, and issuing official certificates or violation reports. The Hotel User interacts with the system primarily to regis ter their establishment and view performance certificates and reports to maintain compliance standards. To foster transparency and public accountability, the Guest actor is empowered to register complaints, upload video evidence of hygiene violations, and review the status of their submissions. By integrating these diverse use cases, the system creates a comprehensive feed back loop that ensures rigorous hygiene standards are maintained through both official auditing and public participation.

The Manage Hotels use case serves as the foundation for the system's multi-tenant architec ture, ensuring that only verified facilities are part of the monitoring ecosystem. Following registration, the Upload Kitchen Video process initiates the core logic of the system, where raw surveillance footage is ingested into the YOLOv8 inference engine for detection. The Review Result use case acts as a human-in-the-loop validation step, ensuring that the AI's detection of pests or PPE violations is accurate before any formal action is taken. Finally, the Generate Certificate/Report use case automates the synthesis of detection data into professional docu mentation, providing hotels with a tangible metric of their sanitary performance.

The Hotel User engages with the system through a streamlined registration workflow de signed to encourage participation in the hygiene program. Once registered, the View Re port/Certificates use case allows management to perform internal audits and display their hy giene ratings to build customer trust. The system boundary extends to the public via the Guest actor, who facilitates decentralized monitoring.

Through the Register Complaints and Upload Video Evidence use cases, guests provide real time feedback that might be missed by fixed surveillance points. The Review use case for guests ensures that their concerns are addressed, creating a closed-loop system where feed back directly informs the hygiene status of the establishment. This tripartite interaction model ensures that the system is not merely a monitoring tool, but a comprehensive platform for hos pitality accountability.

4.3 Activity Diagram

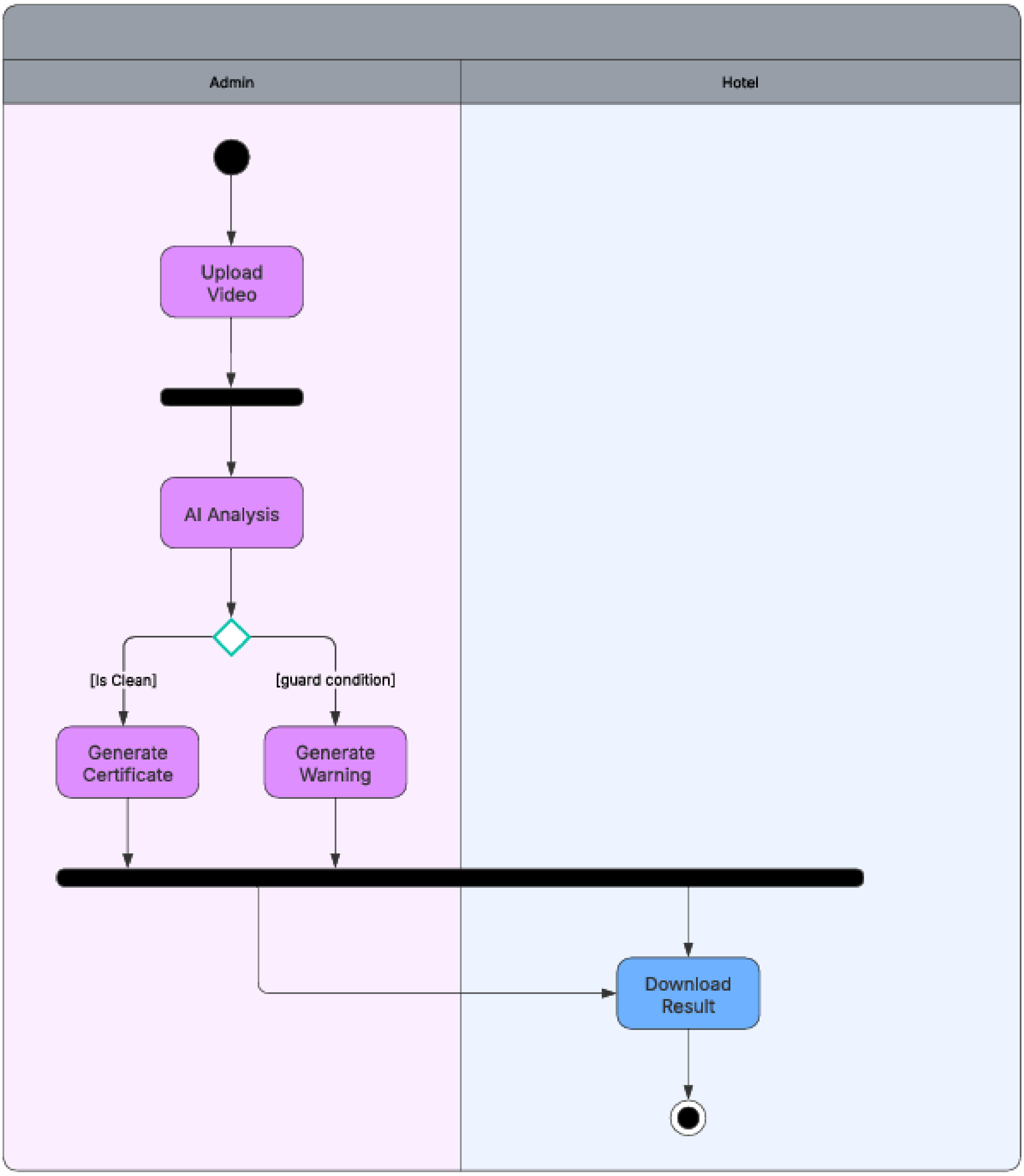

Figure 4.6: Activity Diagram

The Activity Diagram for the AI-Based Hotel Kitchen Cleanliness Monitoring System pro vides a detailed graphical representation of the system's operational workflow, illustrating the sequential flow of control from the initial data ingestion to the final output delivery. The pro cess is logically partitioned into two distinct swimlanes representing the primary stakeholders: the Admin, who oversees the technical processing, and the Hotel, who acts as the consumer of the system's findings. The workflow initiates in the Admin swimlane with the Upload Video activity, where raw surveillance footage from the kitchen environment is transmitted to the cen tralized server. This is followed by a synchronization bar that transitions the control into the core AI Analysis phase. During this stage, the integrated YOLOv8 deep learning model per forms frame-by-frame inference to identify specific hygiene indicators, such as the presence of biological hazards, staff adherence to Personal Protective Equipment (PPE) protocols, and overall environmental sanitation.

The intelligence of the system is encapsulated in the subsequent decision node, which evalu ates the AI's findings against established cleanliness benchmarks. Based on this evaluation, the control flow diverges into two mutually exclusive paths: if the analysis confirms that the facility meets all hygiene standards (the [Is Clean] guard condition), the system triggers the Generate Certificate activity conversely, if the AI detects sanitation lapses or violations (the [Is Dirty] guard condition), it proceeds to Generate Warning. This branching logic ensures that the system handles both positive and negative outcomes with equal efficiency, providing a cus tomized response for every hotel environment. The use of guard conditions within the diagram highlights the system's ability to act as an automated auditor, reducing the need for constant human supervision during the primary screening phase.

Furthermore, the diagram illustrates the synchronization of concurrent processes through the use of fork and join nodes. This architectural choice ensures that the final reporting phase only begins once all AI sub-processes, such as object tracking and class confidence scoring, have reached a state of completion. This prevents the generation of premature or incomplete reports, maintaining the high reliability required for institutional hygiene certification. Once the ad ministrative processing is finalized, the control transitions seamlessly across the organizational boundary into the Hotel swimlane. This represents a hand-off of data where the final Download Result activity is enabled for the end-user. The process terminates at a final state, signifying that the audit cycle is complete and the results have been successfully delivered. This struc tured approach ensures that the complex task of computer vision monitoring is transformed into a predictable, automated, and transparent pipeline, providing immediate and actionable feedback while maintaining a rigorous digital audit trail for public health compliance.

4.4 Sequence Diagram

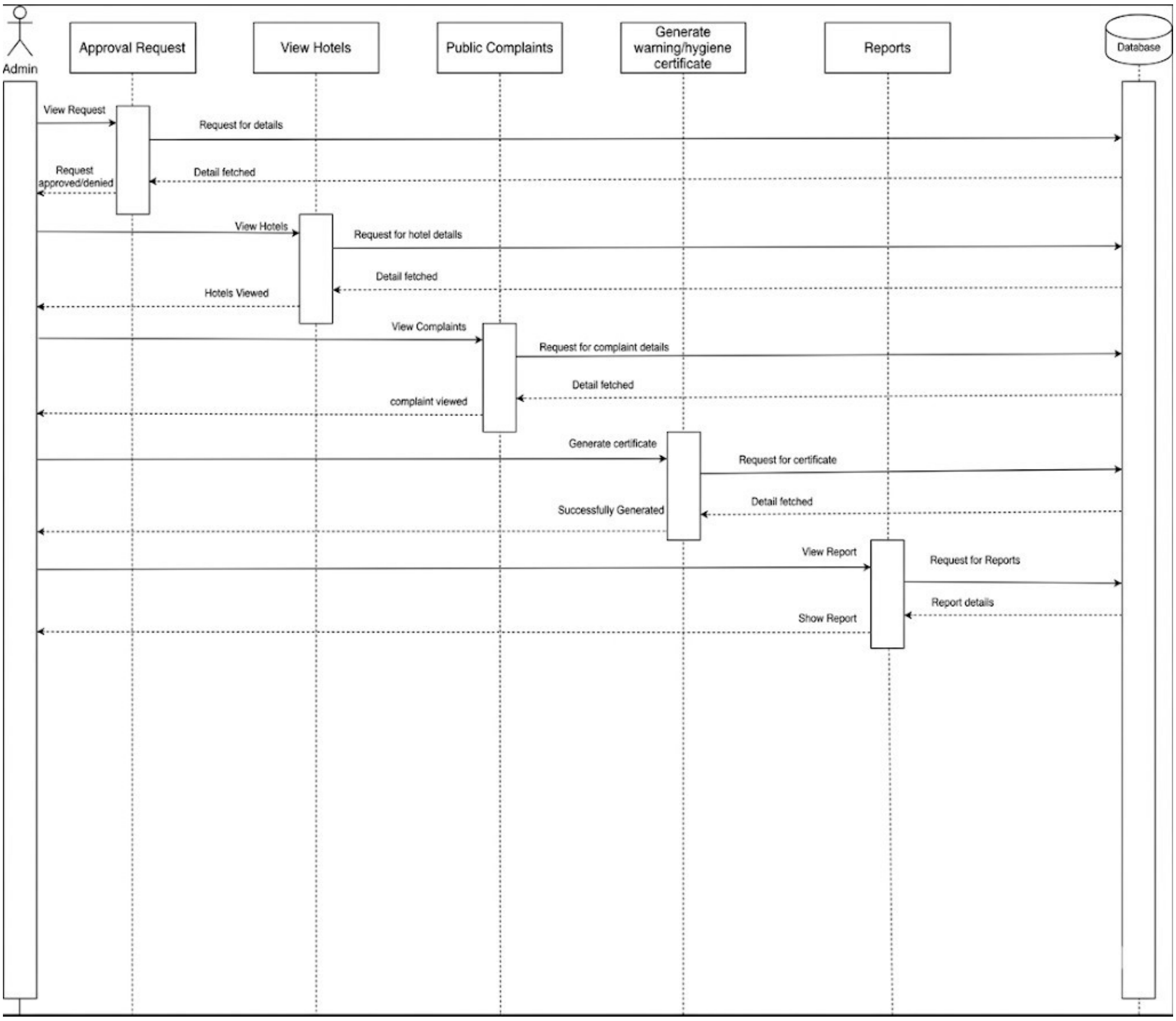

Figure 4.7: Sequence Diagram

The sequence diagram in Figure 4.7 illustrates the chronological interactions between the Ad min actor, various system modules, and the central Database. It highlights how data flows across the platform to complete administrative tasks through a series of request and response cycles.

The process is organized into several key stages:

Approval Management: The Admin initiates a request to view pending hotel approvals. The system fetches the necessary data from the Database, allowing the Admin to either approve or deny the request, with the status updated accordingly.

Hotel and Complaint Oversight: The Admin can view the list of registered hotels and public complaints. For each action, the system sends a request to the Database, which returns the specific details to be displayed on the Admin's interface.

Document Generation: When the Admin triggers the generation of a hygiene certificate or warning, the system requests the relevant audit details from the Database. Once the information is fetched, the document is successfully generated and presented to the Admin.

Reporting: Finally, the Admin can view comprehensive reports. The system queries the Database for report details and renders them back to the Admin for final review.

4.5 List of Identified Classes, Attributes, and Their Relationships

4.5.1 Identified Classes

The Kitchen Hygiene Monitoring System is composed of several core classes that define the system's structure and behavior. These include:

- **User**: The base class for authentication and session management.
- **Hotel Administrator**: Represents the registered establishments responsible for compli ance.
- **Public Guest**: An external actor capable of reporting hygiene concerns.
- **Kitchen Media (CCTV/Image)**: The data container for raw visual evidence.
- **YOLOv8 Inference Engine**: The computational class that executes deep learning mod els.
- **Hygiene Violation Report**: Automates the generation of legal and administrative records.
- **Compliance Certificate**: Validates that a kitchen meets official safety standards.

4.5.2 Identified Attributes

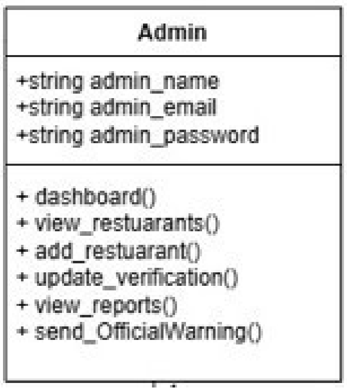

Figure 4.8: Snapshot of Admin Class

The Admin class represents the superuser responsible for overarching system management. It contains credentials such as admin name and admin email. The class provides a comprehensive dashboard() and specialized methods to add restaurants, update verification statuses, and issue official warnings based on hygiene reports.

Figure 4.9: Snapshot of Restaurant Class

The Restaurant class is the core entity for establishment users. It tracks contact details, ver ification timestamps, and the current hygiene status. Functionalities include self-registration, profile management, and the ability to upload media or download generated compliance certifi cates.

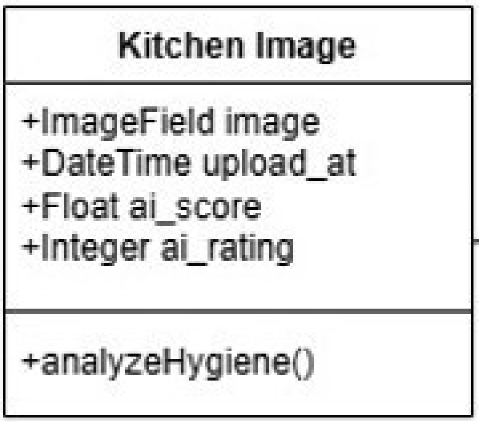

Figure 4.10: Snapshot of Kitchen Image Class

The Kitchen Image class manages individual visual audit records. It stores the ImageField data along with the AI-calculated score and rating. The analyzeHygiene() method serves as the primary trigger for the YOLOv8 inference engine to process the image.

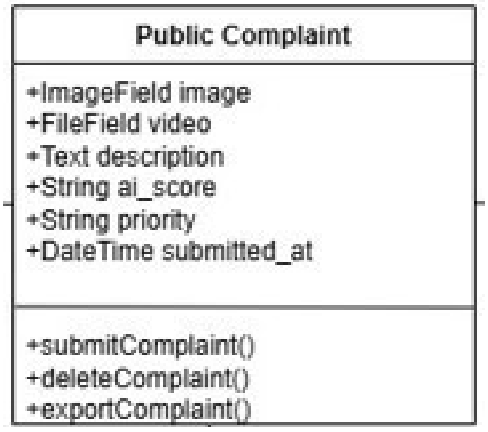

Figure 4.11: Snapshot of Public Complaint Class

The Public Complaint class empowers guests to report hygiene violations. It captures me dia evidence (images/videos) and descriptions, assigning a priority level based on initial AI screening. Methods allow for the submission, deletion, and exporting of complaint data for administrative review.

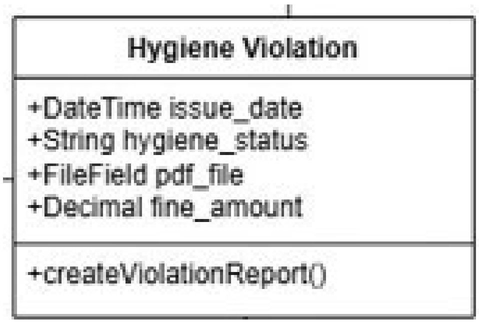

Figure 4.12: Snapshot of hygiene Violation Class

The Hygiene Violation class documents instances where standards are not met. It stores the issue date, the specific status, and the generated PDF report. It also manages the fine amount calculation through the createViolationReport() function.

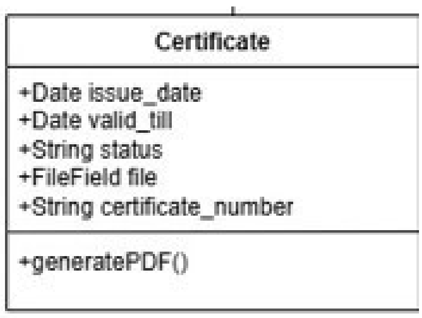

Figure 4.13: Snapshot of Certificate Class

The Certificate class handles the rewarding of clean kitchens. It tracks validity dates and unique certificate numbers. The generatePDF() method utilizes the ReportLab library to create a downloadable, official compliance document for the hotel.

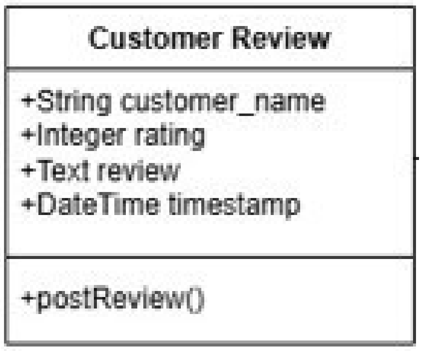

Figure 4.14: Snapshot of Customer Review Class

The Customer Review class provides a feedback mechanism for guests. It records the re viewer's name, star rating, and text feedback with a timestamp. The postReview()method al lows this data to be displayed on the public establishment profile.

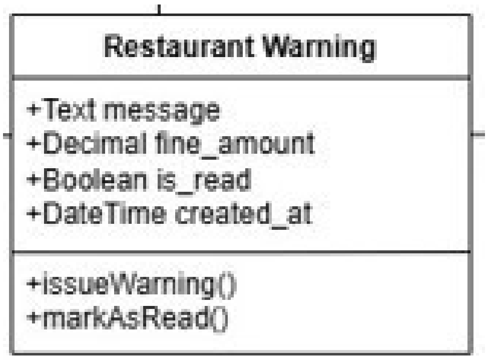

Figure 4.15: Snapshot of Restaurant Warning Class

The Restaurant Warning class is used to notify establishments of pending violations. It includes a specific messageand fine amount. The markAsRead() method allows the system to track whether the establishment has acknowledged the notification.

4.5.3 Identified Classes, Attributes, and Their Relationships

The system architecture is defined by a structured class hierarchy that governs the flow of hy giene data. The system utilizes a One-to-Many relationship between the Hotel Administrator and Kitchen Media, ensuring each establishment can track multiple audit attempts and his torical surveillance logs over time. Furthermore, the Kitchen Media class has a Composition relationship with Detection Results, as detection data and bounding box coordinates are intrin sically tied to the specific frames of the source video and cannot exist independently without an associated media file.

To ensure accountability, the Detection Results class further maintains an Association with a Violation Log entity, which archives specific instances of non-compliance, such as the detection of pests or missing PPE. This log includes attributes for timestamps and class confidence scores, which are vital for generating the final hygiene grade. Additionally, a One-to-One relationship exists between the Analysis Engine and the Report Generator, ensuring that for every completed AI inference cycle, a corresponding digital certificate or warning document is formatted. The Hotel User class is also linked via an Aggregation relationship to the Notification System, allowing the system to push real-time alerts whenever the detection threshold for unsanitary conditions is breached. This interconnected class structure provides a scalable framework that allows for seamless data persistence and rapid retrieval during administrative audits.

Class	**Related Class**	**Association Name**	**Cardinality**
tbl_admin	tbl_hotel	Manages Hotel Accounts	1-*
tbl_hotel	KitchenImage	Uploads kitchen images	1-*
tbl_hotel	UploadModel	Uploads general media	1-*
tbl_hotel	Certificate	Receives certificates	1-*
tbl_hotel	CustomerReview	Receives reviews	1-*
tbl_hotel	PublicComplaint	Receives complaints	1-*
tbl_hotel	HygieneViolation	Receives violations	1-*
tbl_hotel	HotelWarning	Receives warnings	1-*
HygieneViolation	PublicComplaint	Generated from complaint	1-1
HotelWarning	PublicComplaint	References complaint	1-1
HotelWarning	HygieneViolation	References violation	1-1

Table 4.2: Class Association Table

4.6 Class Diagram

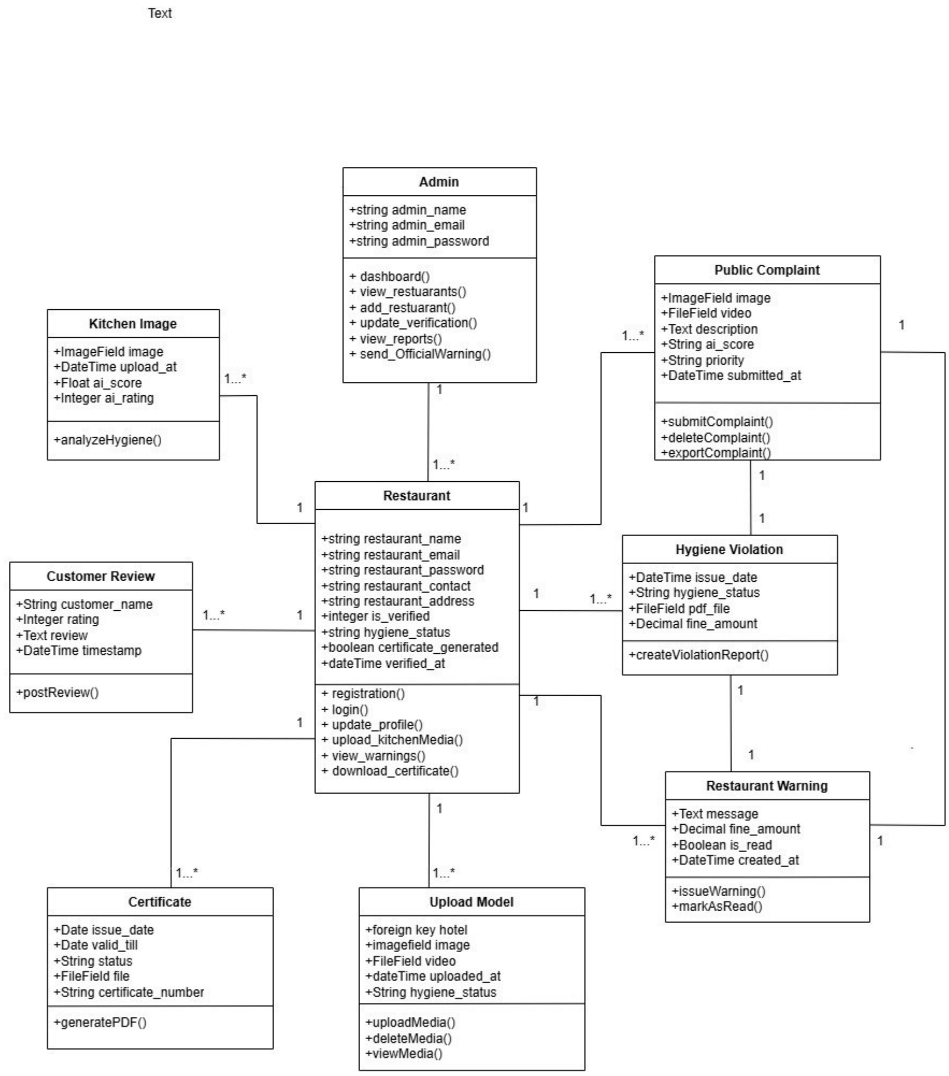

Figure 4.16: Class Diagram

The Class Diagram for the Kitchen Hygiene Monitoring System provides a structural overview of the system by illustrating its classes, attributes, operations, and the relationships between objects. It represents the blueprint of the Django-based architecture, focusing on how data is encapsulated and manipulated.

The central entity is the Restaurant class, which maintains core establishment data and manages functions like registration, media uploads, and certificate downloads. This class serves as the hub for several one-to-many relationships:

Kitchen Image& Upload Model: Handle the storage and AI analysis of surveillance media. Customer Review & Public Complaint: Capture external feedback and evidence from the pub lic, which can trigger higher-priority audits.

Certificate, Hygiene Violation,& Restaurant Warning: Manage the legal and administrative outputs of the system, including PDF generation and fine tracking. The Admin class governs the system at a macro level, with methods to update verification statuses, view global reports, and issue official warnings. This object-oriented structure ensures that the system is modular, allowing for seamless updates to the AI inference logic or database schema without disrupting the core established workflows.

4.7 Database Schema Design

This section provides the detailed attribute specifications for each table, including data types, field sizes, and relational constraints.

1. tbl_admin

Field	**Type**	**Size**	**Constraint**	**Description**
admin_name	Varchar	100	Not Null	Full name of the administrator.
admin_email	Varchar	100	Primary Key	Unique login identifier.
admin_password	Varchar	128	Not Null	Hashed password for security.

Table 4.3: Field Specification for tbl_admin

2. tbl_hotel

Field	**Type**	**Size**	**Constraint**	**Description**
hotel_name	Varchar	100	Not Null	Registered name of the kitchen.
hotel_email	Varchar	100	Unique	Contact and login email.
is_verified	Integer	–	Default 0	0: Pending, 1: Verified, 2: Rejected.
hygiene_status	Varchar	50	Default 'Pending'	Result of the YOLOv8 analysis.

Table 4.4: Field Specification for tbl_hotel

3. PublicComplaint

Field	Type	Size	Constraint	Description
hotel_id	Integer	–	Foreign Key	References tbl_hotel(id).
image	Varchar	255	Not Null	Path to the uploaded evidence.
ai_status	Varchar	50	Not Null	Initial AI assessment.
priority	Varchar	20	Not Null	High/Low based on AI detection.

Table 4.5: Field Specifications for PublicComplaint

4. HygieneViolation

Field	Type	Size	Constraint	Description
fine_amount	Decimal	(10,2)	Not Null	Calculated financial penalty.
pdf_file	Varchar	255	Not Null	Storage path for the PDF report.
issue_date	DateTime	–	Auto Now	Time of violation record creation.

Table 4.6: Field Specification for Hygiene Violation

5. Certificate

Field	Type	Size	Constraint	Description
cert_no	Varchar	50	Unique	Authenticity verification code.
valid_till	Date	–	Not Null	Expiration date of hygiene status.
file	Varchar	255	Not Null	Path to the downloadable certificate.

Table 4.7: Filed Specification For Certificate

6. KitchenImage / UploadModel

Field	**Type**	**Size**	**Constraint**	**Description**
ai_score	Float	–	Range 0–1	Confidence score from YOLO model.
ai_rating	Integer	–	Range 1–5	Normalized cleanliness rating.
image	Varchar	255	Not Null	Media file used for AI inference.

Table 4.8: Field specifications for AI Analysis Tools

7. CustomerReview

Field	**Type**	**Size**	**Constraint**	**Description**
customer_name	Varchar	100	Default 'Anonymous'	Name of the guest providing feedback.
rating	SmallInt	–	Positive	Star rating (1–5) given by user.
review	Text	–	Optional	Qualitative feedback from the guest.
timestamp	DateTime	–	Auto Now	Time when the review was posted.

Table 4.9: Field specifications for CustomerReview

8. HotelWarning

Field	**Type**	**Size**	**Constraint**	**Description**
hotel	ForeignKey	–	tbl_hotel	Links the warning to a specific hotel.
complaint	ForeignKey	–	PublicComplaint	Optional link to the specific public complaint.
violation	ForeignKey	–	HygieneViolation	Optional link to the recorded hygiene violation.
message	Text	–	Not Null	Detailed content of the warning or instruction.
fine_amount	Decimal	(10,2)	Default 0.00	Financial penalty linked to the warning.
is_read	Boolean	–	Default False	Tracks if the hotel owner has read the message.
created_at	DateTime	–	Auto Now	Timestamp of when the warning was issued.

Table 4.10: Field specifications for HotelWarning Table

5 RESULT AND DISCUSSION

The Results and Discussion section provides a comprehensive evaluation of the AI-Based Hotel Kitchen Cleanliness Monitoring System, focusing on the accuracy of the deep learning mod-els and the functional efficiency of the integrated web platform. This chapter analyzes the performance metrics obtained during the testing phase such as precision, recall, and mAP (mean Average Precision) to determine the system's reliability in identifying hygiene vio-lations in real-time. Furthermore, the discussion explores how the automated nature of the system addresses the inherent limitations of manual inspections, highlighting the potential for AI-driven surveillance to significantly enhance public health standards and operational trans-parency within the hospitality industry.

5.1 Overview of Evaluation Metrics

The system was evaluated using Mean Average Precision (mAP), Precision, and Recall. The primary metric for real-world reliability in this hygiene context is mAP@50, which measures accuracy when the predicted bounding box overlaps the ground truth by at least 50 percentage. While Precision was prioritized to minimize false alarms ensuring that a ”clean” kitchen is not wrongly flagged for violations Recall remained crucial for detecting elusive hazards like pests. This balanced metric approach ensures that the system provides a dependable audit trail that can be trusted by both hotel management and health inspectors.

5.2 Training vs. Testing Performance

The custom YOLOv8 model demonstrated excellent generalization capabilities, largely due to the diverse augmentation techniques applied during the preprocessing stage. The performance gap between the training and testing set was only 0.0392, indicating that the model did not over f it and can effectively handle unseen kitchen environments. This high level of stability suggests that the features learned such as the specific textures of kitchen pests and the contours of PPE are robust enough to be deployed across various hospitality facilities with varying lighting and spatial configurations.

```
🏆 OVERALL METRICS (TRAIN)
🟢 Precision:       0.9476
🟢 Recall:          0.9347
🔵 mAP@50:          0.9758 (Standard Accuracy)
🟣 mAP@50-95:       0.7433 (Strict Accuracy)

🔍 CLASS-WISE EVALUATION (TRAIN)
Class Name       | Precision  | Recall     | mAP@50     | mAP@50-95
-----------------------------------------------------------------
apron            | 0.9025     | 0.8405     | 0.9546     | 0.7621
cockroach        | 0.9564     | 0.9411     | 0.9758     | 0.6893
gloves           | 0.9629     | 0.9743     | 0.9909     | 0.7999
hairnet          | 0.9486     | 0.9654     | 0.9810     | 0.6917
lizard           | 0.9810     | 0.9798     | 0.9931     | 0.7910
no_apron         | 0.9314     | 0.9137     | 0.9706     | 0.7521
no_gloves        | 0.9524     | 0.8945     | 0.9648     | 0.7166
no_hairnet       | 0.9734     | 0.9881     | 0.9893     | 0.6960
rat              | 0.9198     | 0.9144     | 0.9622     | 0.7906

============================================================
                 OVERFITTING ANALYSIS
============================================================
Train mAP@50: 0.9758
Test mAP@50:  0.9197
Gap:          0.0561
```

Figure 5.1: Train Results

```
🏆 OVERALL METRICS (TEST)
🟢 Precision:       0.8960
🟢 Recall:          0.8595
🔵 mAP@50:          0.9197 (Standard Accuracy)
🟣 mAP@50-95:       0.6105 (Strict Accuracy)

🔍 CLASS-WISE EVALUATION (TEST)
Class Name       | Precision  | Recall     | mAP@50     | mAP@50-95
-----------------------------------------------------------------
apron            | 0.8859     | 0.8123     | 0.9286     | 0.6793
cockroach        | 0.9462     | 0.9058     | 0.9639     | 0.6366
gloves           | 0.8818     | 0.8718     | 0.9037     | 0.6381
hairnet          | 0.8765     | 0.8819     | 0.9373     | 0.5957
lizard           | 0.9412     | 0.9634     | 0.9767     | 0.6929
no_apron         | 0.8057     | 0.7928     | 0.8832     | 0.5784
no_gloves        | 0.9325     | 0.8872     | 0.9363     | 0.6270
no_hairnet       | 0.9516     | 0.9294     | 0.9743     | 0.6140
rat              | 0.8431     | 0.6907     | 0.7733     | 0.4328
```

Figure 5.2: Test Results

5.3 Precision-Recall (PR) Curve Analysis

The Precision-Recall (PR) curve serves as a vital diagnostic tool for evaluating the performance of the YOLOv8 model, illustrating the intrinsic trade-off between Precision and Recall across various confidence thresholds. In the context of kitchen hygiene monitoring, high precision ensures that the system minimizes false alarms preventing “clean” kitchens from being unfairly penalized while high recall guarantees that the vast majority of actual biological hazards or PPE violations are successfully captured. An ideal model is characterized by a curve that bows toward the top-right corner (1.0,1.0), signifying high accuracy in both positive prediction and target retrieval.

As shown in Figure 6.3, the model demonstrates an exceptional all-classes mAP@0.5 of 0.976, indicating that the system achieves approximately 97.6% accuracy in its combined detection capabilities across the entire dataset. The curve exhibits a desirable “square” shape, where precision remains near the maximum level even as recall increases toward 0.9, proving that the model maintains high reliability even when tasked with detecting nearly every instance in the visual feed. A class-specific breakdown reveals that the system is particularly robust in detecting biological hazards, with the Lizard and Cockroach classes achieving scores of 0.993 and 0.976, respectively. This level of sensitivity is critical for identifying small, fast-moving pests that often evade human inspection.

Similarly, the model shows high competency in enforcing safety protocols, with Gloves (0.991) and Hairnets(0.981) maintaining near-perfect detection rates. Even the negative classes, such as no gloves (0.965) and no apron (0.971), maintain a high Area Under the Curve (AUC), which ensures that administrative warnings are generated with high confidence when non-compliance occurs. Ultimately, the PR curve behavior confirms that the system is well suited for a com mercial hospitality environment, balancing the need for strict hygiene enforcement with the necessity of accurate, data driven reporting.

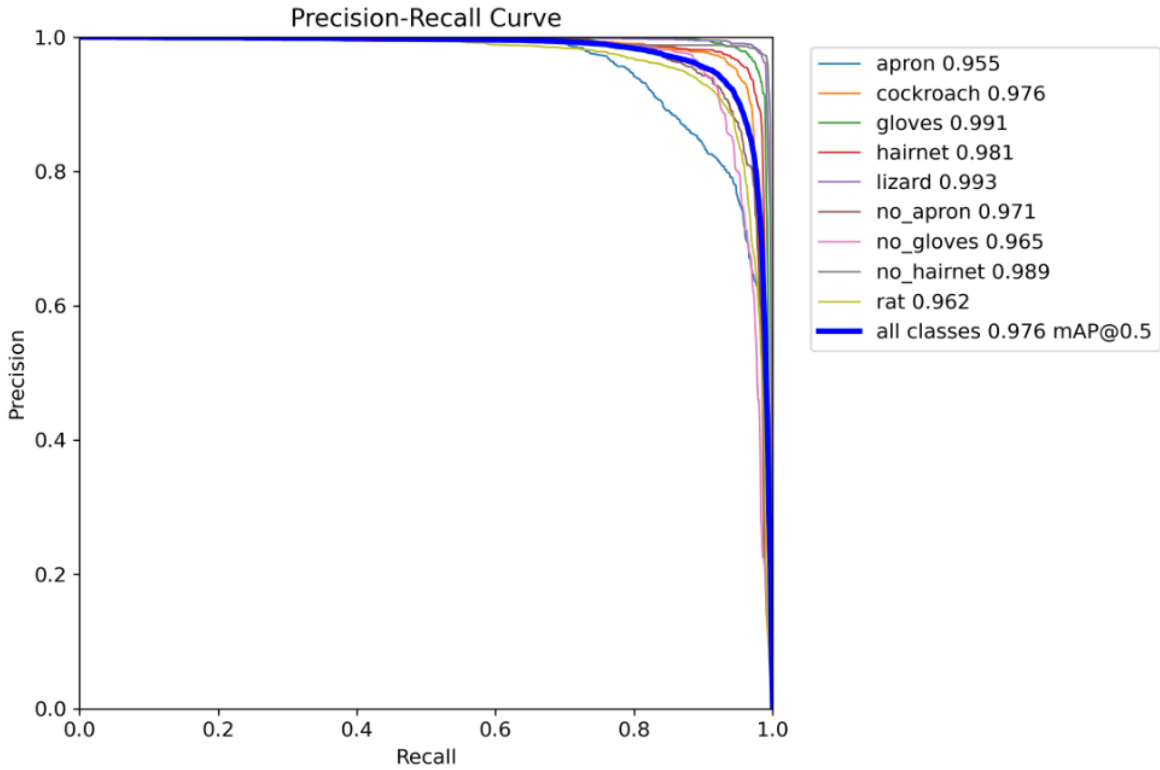

Figure 5.3: Precision-Recall (PR) Curve for Kitchen Hygiene Classes

5.4 F1-Score Curve Analysis

The F1 score is defined as the harmonic mean of Precision and Recall, serving as a singular metric that balances the trade-off between the accuracy of positive predictions and the model's ability to identify all relevant instances. In the context of a kitchen monitoring system, a high F1 score is critical as it ensures the platform maintains low false positives and low false neg atives simultaneously, preventing both unnecessary alarms and overlooked hygiene violations. As illustrated in the F1-Confidence Curve in Figure 6.4, the custom YOLOv8 model achieves a peak overall F1-score of 0.94 at a confidence threshold of 0.440. This specific threshold is recommended for real-world deployment in hotel environments, as it represents the opti mal operating point where the balance between detection sensitivity and reporting accuracy is maximized.

The broad, flat plateau of the curve across the confidence range of 0.2 to 0.7 is a significant in dicator of the model's stability, suggesting that the system provides consistent detection perfor mance even under fluctuating environmental conditions. Furthermore, the high-priority classes specifically Gloves and Lizard maintain superior F1 scores across a wide threshold band, con f irming the system's exceptional robustness in identifying biological hazards and Personal Protective Equipment (PPE) compliance. While the steep decline after the 0.8 threshold is char acteristic of high precision filtering, the overall peak performance confirms that the model is well-tuned for the diverse visual and spatial configurations of a commercial kitchen, ensuring a reliable and automated digital audit trail for hospitality management.

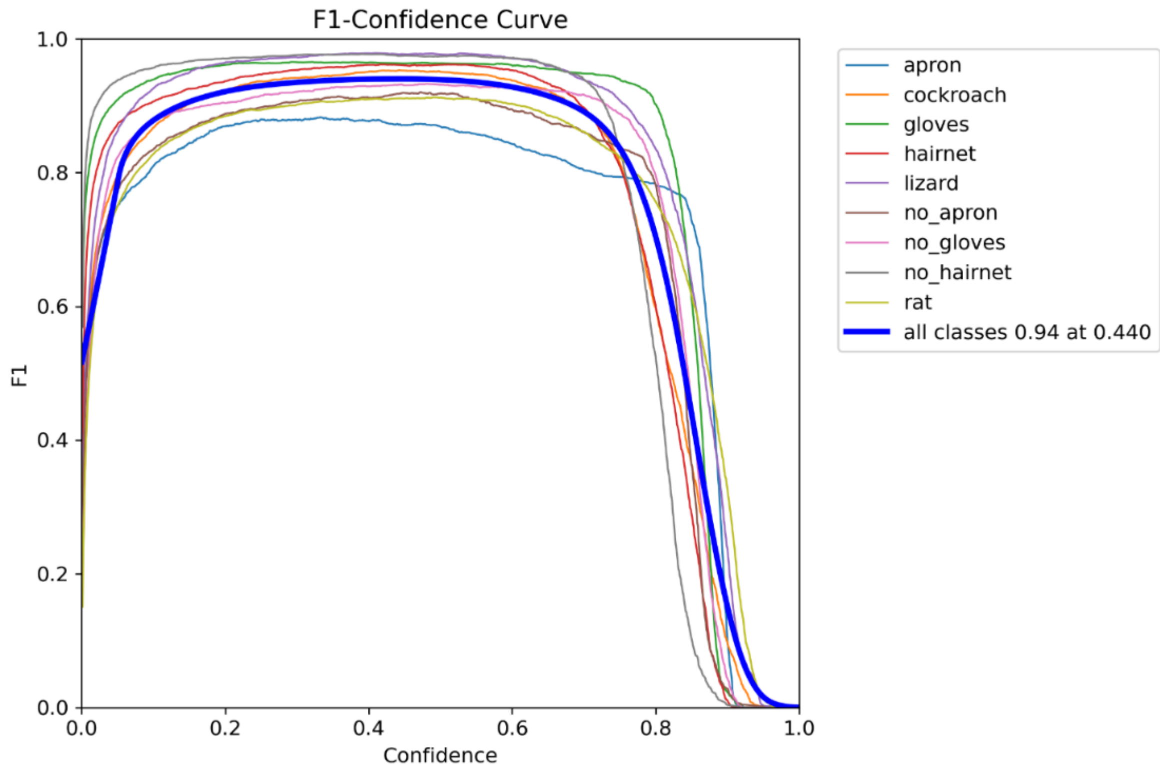

Figure 5.4: F1-Confidence Curve showing optimal threshold for Kitchen Hygiene Detection

5.5 Confusion Matrix Analysis

A Confusion Matrix is a performance measurement tool for machine learning classification problems, presented in a tabular format that compares the actual target values with those pre dicted by the model. It provides a granular view of the model's performance by identifying not only the overall accuracy but also specific areas where the model might be misinterpreting one class for another (False Positives and False Negatives). For the Kitchen Hygiene Mon itoring System, the Normalized Confusion Matrix shown in Figure 6.5 is particularly useful as it expresses these relationships in terms of percentages, allowing for a clear assessment of class-specific reliability.

The diagonal elements of the matrix represent correct predictions, and for this system, they indicate an exceptionally high level of classification accuracy across all categories. Notably, classes such as Lizard(0.99),no hairnet (0.99), Gloves(0.98), and Hairnet (0.98) show near perfect scores, demonstrating the YOLOv8 model's robust ability to identify biological hazards and PPE compliance. Even the most complex classes, such as Rat (0.95) and no apron (0.95), maintain high precision. The minor leakage seen in the "Background" column suggests that some small objects are occasionally missed (False Negatives), which is a common occurrence in high-resolution kitchen surveillance due to motion blur or occlusion.

However, the negligi ble values in the off-diagonal cells prove that the model rarely confuses one active class with another, ensuring that the hygiene reports and violation warnings generated by the system are highly credible and data-driven.

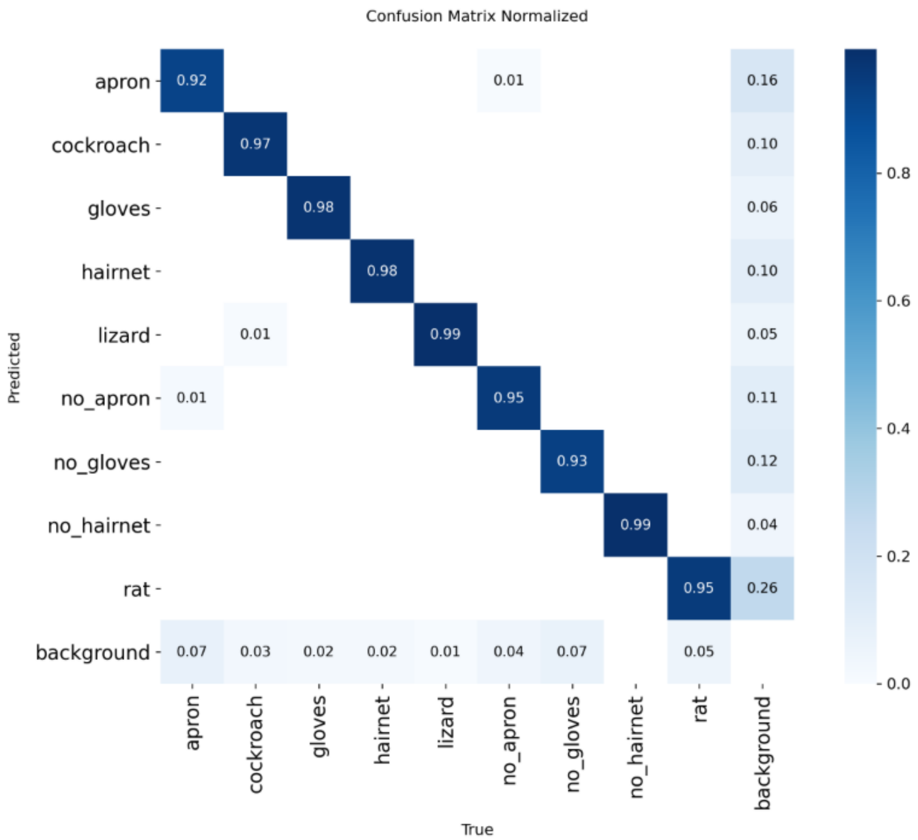

Figure 5.5: Normalized Confusion Matrix for the YOLOv8 Hygiene Detection Model

5.6 Prediction Statistics and The ‘Augmentation Efficacy

Prior to data balancing, the rat class suffered from a severe minority class imbalance, yielding poor early detection scores (mAP@50   0.38). By implementing a targeted Rat preprocessing pipeline using the Albumentations library, the prediction statistics improved dramatically. This pipeline focused on simulating the unique challenges of detecting small, dark objects against the complex backgrounds of a kitchen floor. The improvement can be quantified through the following key metrics:

- Onthe final unseen test set, the rat class achieved anmAP@50 of 0.773 (77.3%).
- Onthe training set, the rat class reached an mAP@50 of 0.928 (92.8%).

While rats remain the mostchallenging class to perfectly box as evidenced by the strict mAP@50 95 score of 0.420 the system is now highly capable of triggering accurate presence alerts. The success of this approach is attributed to the simulation of low light CCTV conditions, motion blur, and varied camera angles, which trained the model to recognize the specific contours of pests even in grainy footage. This targeted data engineering bypassed the need for capturing thousands of rare real-world pest sightings, effectively solving the data scarcity problem com mon in high-priority safety audits. Furthermore, the integration of CIoU loss during training ensured that even with small bounding boxes, the localization remained stable enough to pre vent flickering detections. This efficacy proves that for specialized AI applications, the quality and targeted nature of the training data often outweigh the sheer volume of generic images.

5.7 Findings and Conclusion

The results confirm that the custom 9-class YOLOv8 architecture successfully fulfills the ob jectives of the automated Kitchen Hygiene monitoring system. The evaluation proves that the model is robust enough to act as an objective auditor in commercial environments.

- **High Reliability:** With an overall test mAP@50 of over 92%, the system is production ready for triggering real-time Smart Alerts for hygiene violations.
- **Robust PPE Logic**: Near-perfect scores (95%+) in the confusion matrix for aprons, hairnets, and gloves prove the system can reliably automate daily compliance auditing without human bias.
- **Successful Data Engineering**: The project proves that heavily targeted, domain-specific data augmentation (e.g., adding noise to simulate grainy security cameras) is highly ef fective for solving class imbalances in pest detection.

The seamless integration of the YOLOv8 inference engine with a Django-based web frame work demonstrates that complex deep learning models can be made highly accessible for non technical hotel staff through intuitive dashboards. By automating the generation of compliance certificates and violation reports, the system transforms raw surveillance data into actionable administrative records, significantly reducing the workload of manual health inspectors. Fur thermore, the dual-actor ecosystem, allowing both administrators and the public to participate in hygiene monitoring, fosters a new level of transparency and accountability within the hospi tality sector.

This project establishes a foundational framework for Smart Kitchens, where artificial intel ligence acts as a continuous, objective guardian of public health. Ultimately, the successful deployment of this monitoring system proves that AI-driven visual auditing is not just a theo retical concept, but a viable, high-performance solution for modernizing food safety standards globally. Beyond mere detection, the system's ability to store historical data in a secure SQLite database allows establishments to track their hygiene trends over months, enabling a shift from reactive cleaning to predictive maintenance.

The implementation further highlights that the modularity of the YOLOv8 Django stack al lows for rapid scaling to include emerging sanitary threats, such as detecting improper waste disposal or cross-contamination behaviors. By utilizing edge-compatible weights, the system ensures that high-speed inference can be maintained without the need for expensive high-end server infrastructure, making it economically feasible for small-scale restaurants. The integra tion of the Augmentation pipeline specifically for rats validates that synthetic data manipulation can bridge the gap in rare-event detection, a major hurdle in previous automated surveillance research. Furthermore, the system effectively acts as a deterrent, as the mere presence of an automated AI auditor encourages kitchen staff to maintain consistent PPE compliance through out their shifts. This creates a self-sustaining culture of safety where the AI provides not just oversight, but continuous reinforcement of hygiene best practices. As the system scales to han dle live video streaming, it will evolve into a real-time intervention tool capable of preventing contamination before food reaches the consumer. In conclusion, the system bridges the gap be tween passive video surveillance and active, intelligent hygiene management, offering a highly accurate, scalable solution for the modern food service industry.

6 TESTING

Software testing is a critical phase in the Kitchen Hygiene and Pest Detection System devel opment process, ensuring that the system operates as intended before real-world deployment. It serves as the final review of the system's specifications, design, and model accuracy, ver ifying that the software meets technical and user requirements while being free from errors. To ensure the system functions correctly, tests must be conducted systematically using well structured test cases designed with discipline and precision. The testing process aims not only at finding faults in the existing software but also at finding measures to improve the software in terms of detection efficiency, bounding-box accuracy, and system usability.

6.1 Unit Testing

Unit testing for the Kitchen Hygiene System focuses on verifying the fundamental modules in isolation to ensure each component functions as intended before full system integration. The process begins with the authentication module, where tests validate that the three distinct roles Admin, User, and Guest can securely access their specific dashboards and that unauthorized login attempts are correctly rejected. The media upload module is tested to confirm it correctly processes diverse CCTV file formats and successfully applies the 640×640 pixel resizing required for the deep learning pipeline. For the AI inference module, unit tests are performed on the YOLOv8 engine to ensure it accurately localizes and classifies the nine target classes, including PPE compliance and biological hazards like rats or cockroaches.

Furthermore, the hygiene scoring logic is verified independently to ensure that raw detection data is accurately converted into a normalized 5-star rating for the user interface. The certificate generation module, powered by ReportLab, is tested to ensure it can dynamically pull results from the SQLite database to produce professional PDF documents without errors. To ensure overall reliability, control path testing is utilized to track the logical flow of data, while boundary condition testing confirms the system remains stable even when handling extreme inputs or empty data fields. Finally, database unit tests ensure that all violation logs and certificate metadata are consistently archived and retrieved from the cloud-hosted infrastructure on AWS EC2.

Sl.no	Procedure	Expected Output	Actual Output	Status
1	Hotel Registration with valid details	Registration Successful	Registration Successful	Pass
2	Login with incorrect credentials	Invalid Login	Invalid Login	Pass
3	Upload CCTV Image/Video	Media successfully uploaded	Media successfully uploaded	Pass
4	Upload unsupported file format	Error message	Error message	Pass
5	YOLOv8 inference on image with PPE	AI detects PPE	AI detects PPE	Pass

Table 6.1: Unit Test Cases

6.2 Integration Testing

Integration testing for the Kitchen Hygiene System focuses on the seamless communication between interconnected modules to ensure the entire monitoring pipeline operates reliably. A critical verification point is the model-database synchronization, which ensures that detection arrays from the YOLOv8 engine are accurately mapped and written to the SQLite database without data loss. The media pipeline is also tested to confirm that once a user uploads an image, it is correctly preprocessed and ingested by the AI model for real-time analysis. To maintain system security, integration checks verify that the Admin dashboard accurately aggregates reports from diverse Hotel Owner accounts while maintaining role-based data isolation. Furthermore, tests validate the interaction between the database and the ReportLab engine, ensuring the Generate Certificate action injects the correct session-specific metadata into the final PDF. The system's responsiveness is verified by testing the real-time alert trigger. Finally, the integration between the frontend and the AWS EC2 cloud infrastructure is evaluated to ensure that establishment hygiene scores and financial metrics, such as outstanding fines, update dynamically across all user interfaces.

Testing also confirms that any change in an establishment's hygiene status automatically updates the Live Dashboard and the Kitchen Reports table simultaneously to maintain data consistency across all user views. The integration suite further validates that when a certificate is deactivated in the administrative backend, the Download PDF button is immediately disabled on the hotel owner's management screen to prevent the circulation of invalid credentials. Additionally, the system's ability to handle high concurrency is evaluated by simulating multiple simultaneous media uploads from different hotels to ensure the cloud-hosted YOLOv8 engine and database can manage parallel processing without service degradation. Moreover, the connection between the detection module and the alert system is rigorously tested to guarantee that high-priority biological hazards, such as rats or cockroaches, trigger immediate notifications via the Firebase messaging service. Finally, the workflow between the user's manual image review and the automated database archival is synchronized to ensure every audit trail remains transparent and tamper-proof for public health inspections.

6.2.1 Integration Test Cases

Sl.no	Procedure	Expected Output	Actual Output	Status
1	Hotel uploads clean kitchen video → AI analyzes → DB updates status	Status updated to "Clean" in database	Status updated to "Clean" in database	Pass
2	Guest uploads evidence of pests → AI analyzes → Admin dashboard updates	Status updated to "Dirty" and complaint logged	Status updated to "Dirty" and complaint logged	Pass
3	Request certificate for a "Clean" kitchen	Auto-generated certificate downloads successfully	Auto-generated certificate downloads successfully	Pass
4	Request certificate for a "Dirty" kitchen	Unauthorized access / Request denied	Unauthorized access / Request denied	Pass
5	System detects no hairnet and no gloves → Generates violation	Violation report generated detailing missing PPE	Violation report generated detailing missing PPE	Pass

Table 6.2: Integration Test Cases Table

6.3 System Testing

System testing is the final phase in the testing process, where the Kitchen Hygiene System is tested as a complete product to ensure it meets all functional, security, and performance require ments. This phase involves testing the entire system with structured and planned test data (such as unseen, real-world CCTV footage) to validate its behavior under real-world conditions. This phase includes both verification and validation, categorized into two methods:

- **Static Testing**: This involves reviewing the system's code and documentation without executing it, to identify any inconsistencies, missing requirements, or logical errors.

- **Dynamic Testing**: This involves executing the system to observe its behavior and vali date that it meets the specified requirements.

During system testing, all the modules of the Kitchen Hygiene System are tested together to ensure smooth overall functionality. Tests are conducted to verify:

- Login and role-based access management for different users (Admin, Hotel, Guest).
- The end-to-end pipeline of media upload, real-time AI processing, and result display.
- The accuracy of the YOLOv8 model on unseen test data (ensuring mAP@50 remains above 90%).
- Automated status classification ("Clean" vs. "Dirty") based on detection confidence thresholds.
- The seamless generation and downloading of PDF compliance certificates and violation reports.

The testing process ensures that the Kitchen Hygiene System operates efficiently, securely, and reliably, delivering a seamless automated monitoring experience for the food service industry.

Sl.no	**Procedure**	**Expected Output**	**Actual Output**	**Status**
1	Hotel uploads clean video $\rightarrow$ AI $\rightarrow$ DB	Status updated to "Clean"	Status updated to "Clean"	Pass
2	Guest uploads pest evidence $\rightarrow$ Admin	Status updated to "Dirty"	Status updated to "Dirty"	Pass
3	Request certificate for "Clean" kitchen	Certificate downloads	Certificate downloads	Pass
4	Request certificate for "Dirty" kitchen	Request denied	Request denied	Pass

Table 6.3: Integration Test Cases

7 MODEL DEPLOYMENT

The deployment phase focuses on transitioning the trained YOLOv8 engine into a production-ready service capable of providing real-time hygiene analysis. This process involves integrating the deep learning weights with the Django backend and hosting the entire architecture on a scalable cloud environment to ensure accessibility and reliability for hospitality administrators.

The model implementation phase begins with the integration of the trained YOLOv8 engine into the core Python-based backend, utilizing the PyTorch (.pt) weight file for high-speed inference. When a user interacts with the system, the implementation script automatically initiates a preprocessing pipeline that resizes all incoming kitchen media to a standardized 640×640 pixel resolution to ensure compatibility with the deep learning model's input layers. The model utilizes an anchor-free detection mechanism to localize and classify nine specific target classes, including PPE compliance markers and biological hazards like pests. To ensure reliable administrative oversight, the inference logic is configured with an 89% precision threshold, which filters out low-confidence detections and ensures that only verified hygiene violations are logged. Once the analysis is complete, the detection arrays and confidence scores are serialized and passed to the SQLite database, while the ReportLab library dynamically pulls this data to generate official PDF certificates for compliant establishments.

Beyond the initial inference, the implementation incorporates a Non-Maximum Suppression (NMS) layer to refine overlapping bounding boxes, ensuring that each hygiene violation or PPE item is counted only once per frame. The system utilizes a multi-threaded execution approach, allowing the Django backend to remain responsive while the YOLOv8 engine processes high-resolution media in a separate worker process. For visual transparency, the deployment script uses OpenCV to dynamically draw color-coded bounding boxes green for compliance and red for violations directly onto the output media. These processed frames are then converted into a base64 encoded format for immediate rendering on the user's Live Dashboard. To maintain high operational standards, the implementation includes a logging utility that tracks inference latency, ensuring that every kitchen scan is completed within a sub-second timeframe. The backend logic is further optimized to perform periodic database cleanups, archiving older detection metadata into long-term storage to prevent SQLite performance degradation. Furthermore, the implementation ensures that the model's confidence scores are weighted against the specific class; for instance, biological hazards like pests are prioritized with a lower detection threshold to maximize safety sensitivity. The connection between the inference engine and the ReportLab module is managed through a secure API call, ensuring that only verified "Clean" verdicts can trigger the certificate generation sequence. Finally, the implementation includes a fail-safe mechanism that reverts to a secondary General Hygiene classification if the input image quality is too low for specific PPE detection.

The cloud implementation of the system is architected on the AWS EC2 platform, utilizing the Ubuntu 24.04 LTS (Noble Numbat) operating system to provide a secure, long-term stable environment for the deployment. The system specifically leverages the m7i.large instance type, featuring 7th Generation Intel Xeon Scalable processors, which offers the computational efficiency required to run the YOLOv8 engine and the Django backend concurrently without significant latency. To support the project environment, deep learning model weights, and the expanding database of hygiene reports, the instance is provisioned with 25 GB of General Purpose (gp3) Elastic Block Store (EBS) storage. Connectivity is maintained through the association of an Elastic IP, ensuring that the hosted web application remains reachable via a persistent public address even after system reboots or maintenance cycles.

To ensure robust network security, the instance is protected by a tailored Security Group with strict Inbound Rules. Access is restricted to essential services, including Port 22 (SSH) for secure administrative terminal access, Port 80 (HTTP) and Port 443 (HTTPS) for standard web traffic, and Port 8000 for the Django development and testing environment. All other ports are closed by default to minimize the attack surface. Outbound rules are configured to allow all traffic, enabling the system to communicate with external services like Firebase for real-time alerts and GitHub for version control updates. This robust cloud setup ensures that the Kitchen Hygiene Monitoring System can handle real-time media processing and data archival with high availability, providing a professional and stable platform for administrative oversight in the hospitality sector.

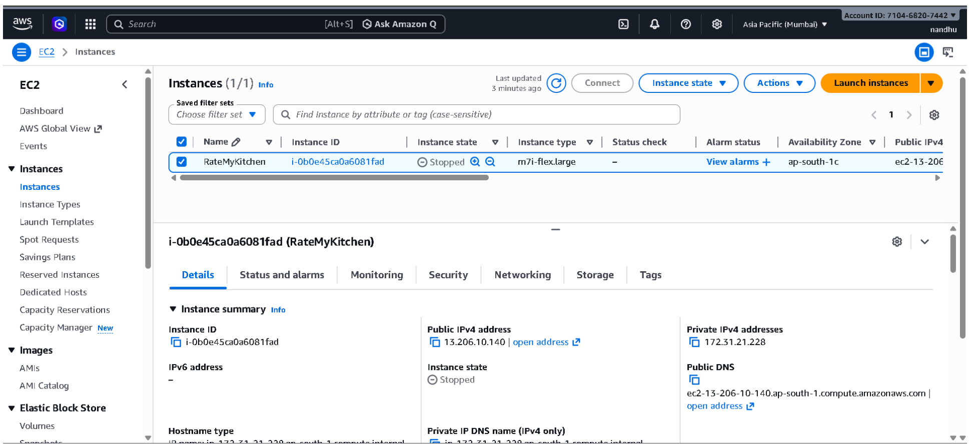

Figure 7.1: Aws Console

7.1 GUI Design

The Graphical User Interface (GUI) is a critical component of the AI-Based Hotel Kitchen Cleanliness Monitoring System. Developed using HTML, CSS, and Bootstrap within the Django framework, the interface is designed to be highly intuitive and responsive. It ensures that administrators, hotel staff, and guests can easily navigate the system, upload media, and interpret the AI-generated hygiene reports without requiring specialized technical knowledge.

Admin Dashboard

The Admin Dashboard serves as the central control panel. It allows authorized personnel to monitor overall compliance across different hotels, view public complaints, and manage system users.

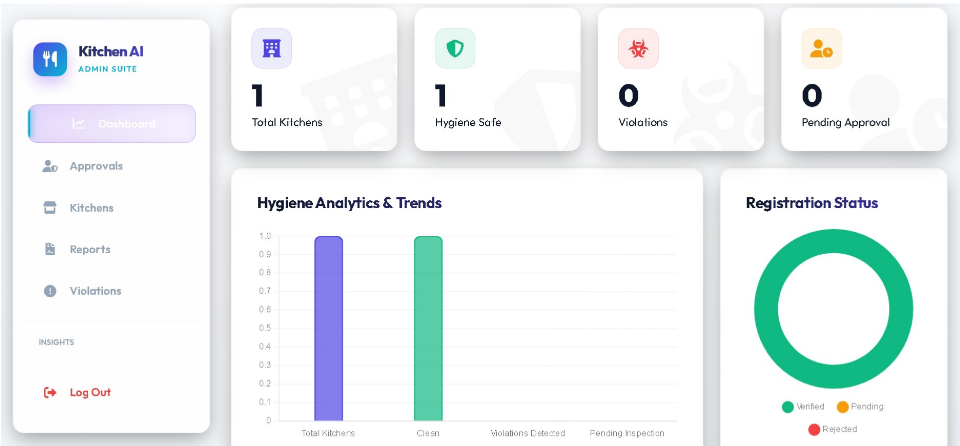

Figure 7.2: Admin Dashboard Interface

Kitchen Media Upload Interface

This interface is designed for hotel staff and guests to submit visual data. Users can securely upload images or CCTV footage of the kitchen environment, which is then sent directly to the YOLOv8 deep learning model for automated analysis.

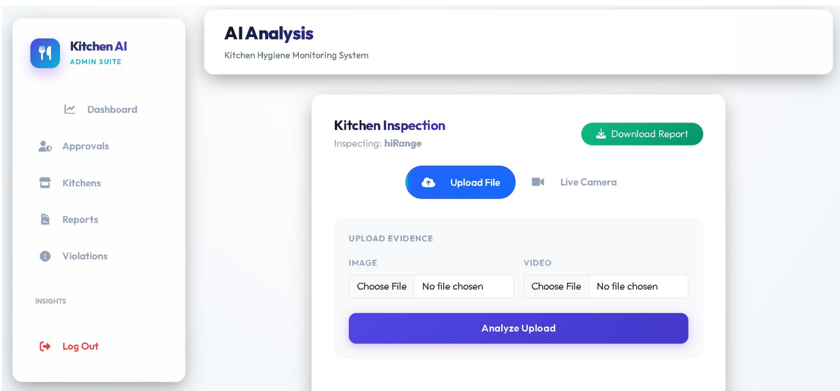

Figure 7.3: Media Upload Screen

AI Hygiene Analysis Results

The AI Hygiene Analysis Results screen serves as the final stage of the automated monitoring pipeline, dynamically rendering the AI verdict after YOLOv8 engine processing. For compliant kitchens, the status is labeled ”Clean” based on the verified absence of biological hazards and high PPE confidence scores. To ensure transparency, the system overlays detection bounding boxes onto the source media for administrative verification. For these audits, users can instantly download an official PDF Compliance Certificate.

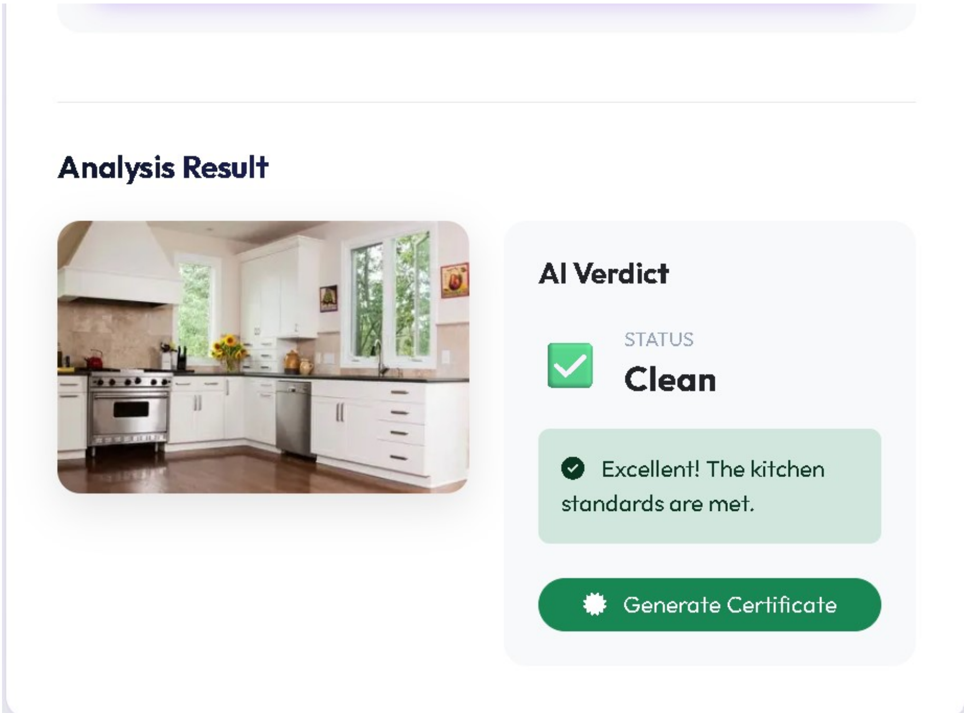

Figure 7.4: Media Upload Screen

Reports

The Hygiene Reports interface, shown in Figure 5.28, is a centralized hub for monitoring cleanliness standards across all registered establishments. The reports table displays hotel names, contact emails, and current AI-evaluated hygiene statuses for quick administrative oversight. Administrators can utilize search features and status filters to track compliance trends and identify hotels requiring corrective action. Ultimately, this interface transforms raw AI detection data into organized, actionable records for the hospitality sector.

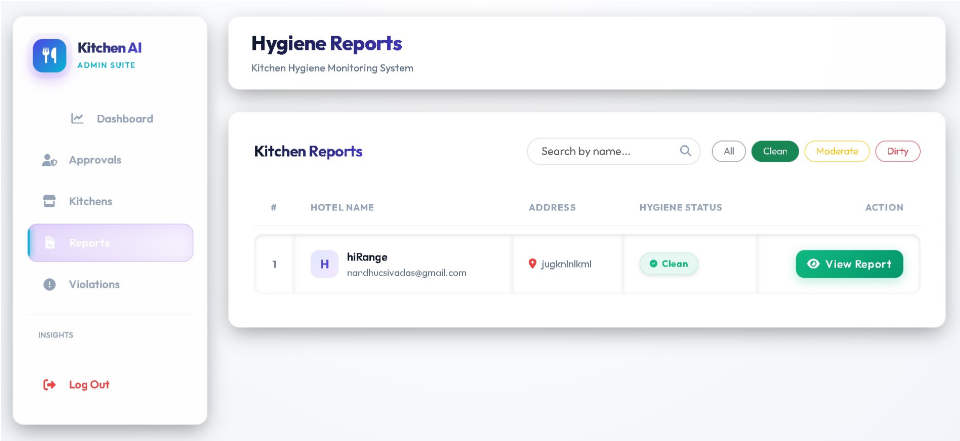

Figure 7.5: Report Screen

Violations and Citizen Reports Dashboard

The Violations dashboard is dedicated to managing public complaints and tracked hygiene infractions. It features high-level summary cards that immediately alert administrators to the total number of reports and pending items requiring action.

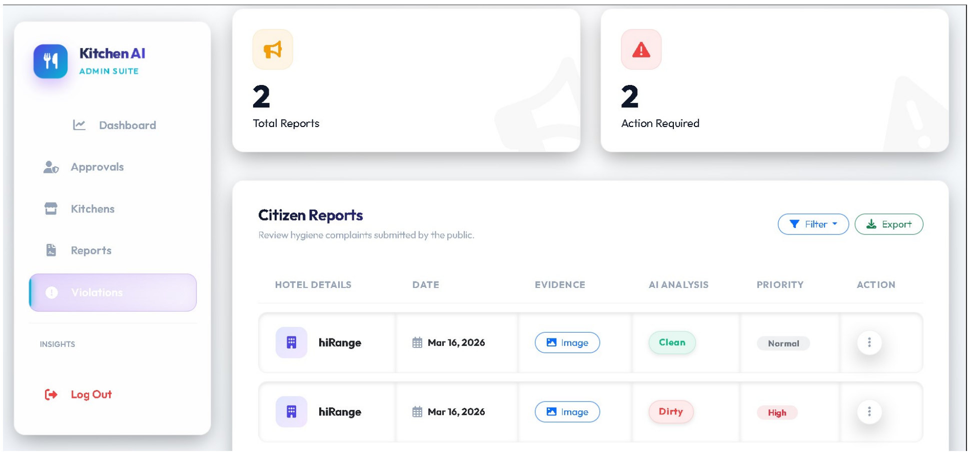

Figure 7.6: Violations and Citizen Reports Interface

Hotel Owner Dashboard

The Hotel Owner Dashboard offers a personalized, real-time summary for restaurant managers to monitor establishment compliance and hygiene metrics. A centralized Current Status Overview panel displays the kitchen's AI-evaluated status alongside totals for earned certificates, kitchen scans, and customer reviews. The interface also tracks Outstanding Fines to notify management of any pending penalties resulting from detected hygiene violations. Primary call-to-action buttons for My Certificates and View Reports provide direct access to official documentation and detailed diagnostic findings. Built with a responsive, dark-themed layout, this dashboard empowers owners to maintain proactive, data-driven food safety standards.

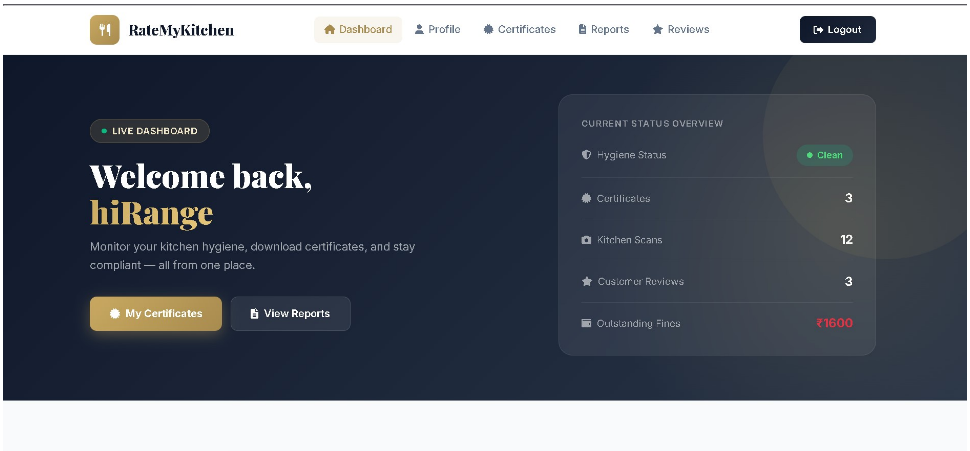

Figure 7.7: Hotel Owner Dashboard Interface

Certificate Management Interface

The Certificate Management Interface provides a specialized portal for hotel owners to track and manage their official hygiene compliance credentials in a clear historical record. The system distinguishes between Active and Deactivated certificates, each displaying essential metadata such as unique certificate numbers and validity periods. For active credentials, users can obtain professional documents via the Download PDF button, while access is restricted for expired or deactivated awards. A secondary tab for ”Violation Alerts” allows managers to quickly switch between viewing their safety rewards and pending infractions. Ultimately, this interface maintains a transparent audit trail that validates an establishment’s adherence to rigorous hygiene protocols through data-driven AI verification.

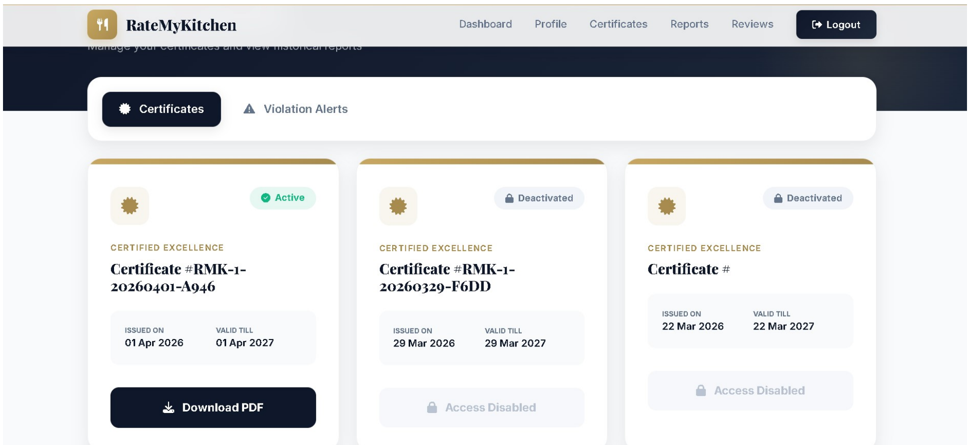

Figure 7.8: Certificate Management and Download Screen

Inspection Reports History

The Inspection Reports History interface provides hotel management with a comprehensive chronological log of all AI-powered hygiene assessments performed over time. This specific screen acts as a critical accountability tool, explicitly highlighting instances where the system identified sanitary lapses, as indicated by the Dirty Detected status labels. Each entry in the historical table meticulously records the exact Inspection Date, allowing facility managers to track their performance across different shifts and weeks. To facilitate deeper analysis, a View Full Report button is provided for every entry, granting immediate access to the detailed di agnostic breakdown generated by the YOLOv8 engine. This feature is essential for helping restaurant owners identify recurring bottlenecks in their hygiene protocols and understand spe cific areas of non-compliance. By maintaining this digital audit trail, the system ensures that management can take data-driven corrective actions to improve future hygiene scores. The in terface is designed with a clean, professional aesthetic using Bootstrap, ensuring that complex inspection data is easy to read at a glance. Integration with the secure SQLite database ensures that these records remain immutable and accessible for official health audits. Furthermore, the responsive layout allows management to review their establishment's cleanliness history from any mobile or desktop device. Ultimately, this screen bridges the gap between passive surveil lance and active operational management, fostering a culture of continuous improvement in food safety.

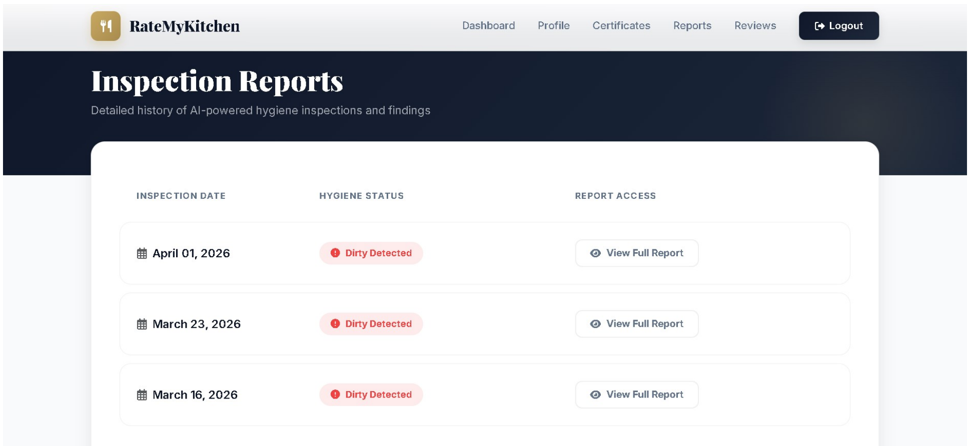

Figure 7.9: Inspection Reports History Screen

Guest Feedback Interface

The Guest Feedback and Ratings Dashboard, is a dedicated inter face designed to provide hotel management with real-time insights into customer satisfaction and public perception. The dashboard features a prominent Average Rating display, currently showing a 4.0-star score based on guest reviews, providing an immediate snapshot of the es tablishment's general standing. To offer a more nuanced understanding of guest experiences, the GUI includes specific sentiment metrics, such as Service Quality and Hygiene Sentiment, represented by color-coded progress bars. Below the aggregate scores, individual reviews from anonymous guests are listed chronologically, featuring their specific star ratings and textual comments like "the food was delicious" or "nice food". This level of detail allows restaurant owners to correlate subjective public sentiment with objective, AI-driven hygiene compliance scores. The navigation bar at the top ensures that managers can quickly switch between this feedback loop and other critical modules like Certificates or Reports. Developed with Boot strap, the interface is fully responsive, allowing management to monitor guest interactions from mobile or desktop devices at any time. By consolidating this data, the system promotes trans parency and helps establishments identify specific service areas that may require improvement. Ultimately, this interface serves as a vital tool for maintaining high standards of customer ser vice while reinforcing the establishment's commitment to transparency

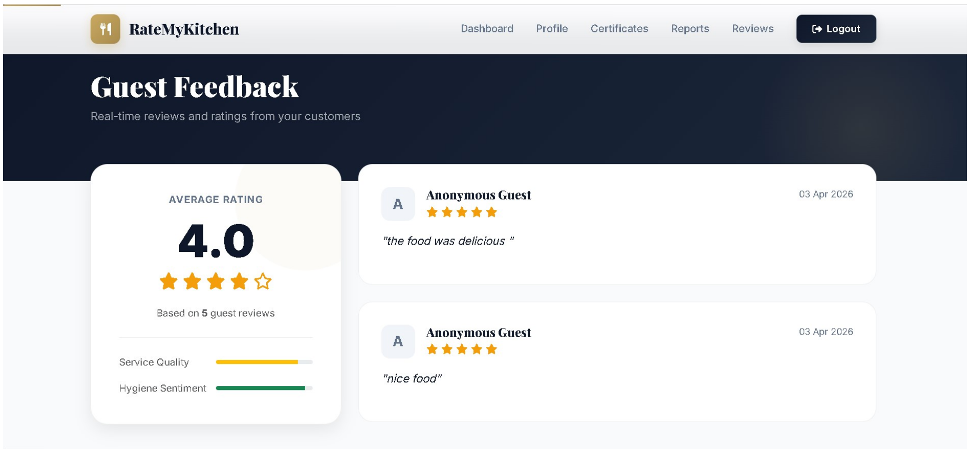

Figure 7.10: Guest Feedback and Ratings Dashboard

Hotel Profile Management

The Hotel Profile Management interface allows establishment owners to view and update their registered business details within the RateMyKitchen platform. The screen displays essential contact information, the physical address of the facility, and a quick-view badge indicating the current overarching hygiene compliance status. Facility managers can seamlessly edit their pro f ile data to ensure regulatory communication remains accurate, and they can navigate directly to their Certification History to track their earned compliance badges over time.

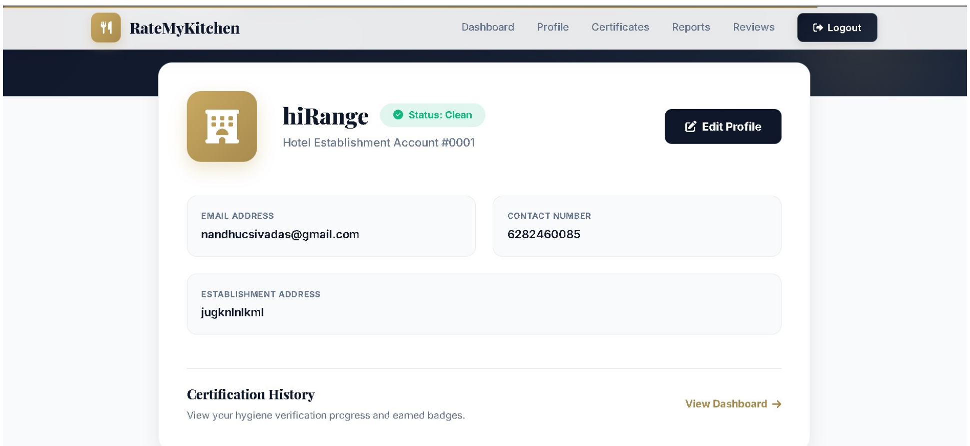

Figure 7.11: Hotel Profile Management Screen

8 GIT HISTORY

The Git History section documents the evolutionary development of the Kitchen Hygiene System, showcasing the systematic approach taken during the project's lifecycle. By utilizing GitHub as a version control platform, the development process was maintained through a series of structured commits that track feature implementation, bug fixes, and architectural refinements.

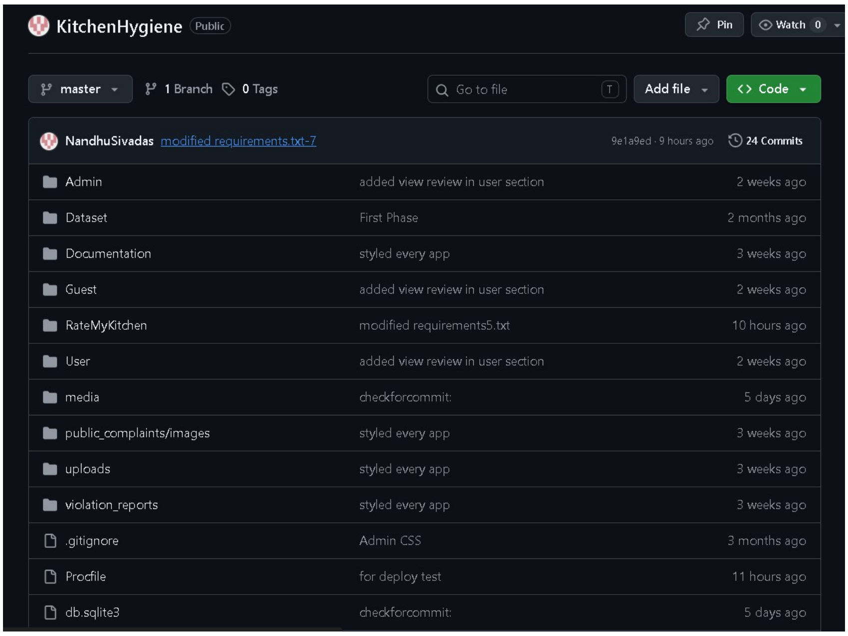

Figure 8.1: Git History

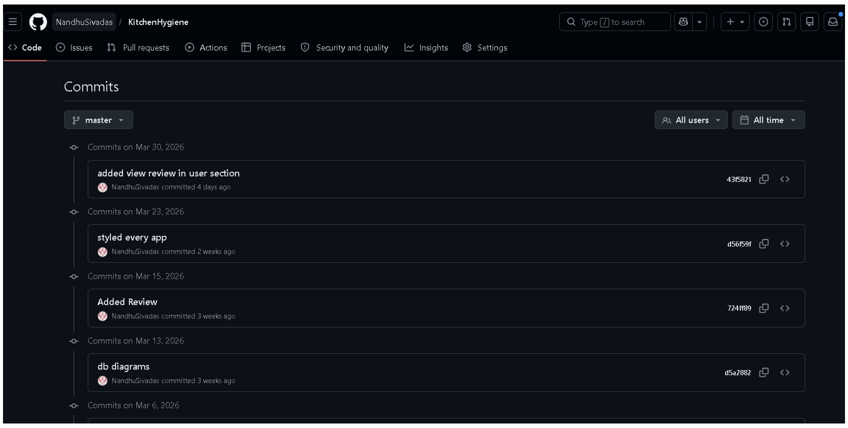

Figure 8.2: Git History

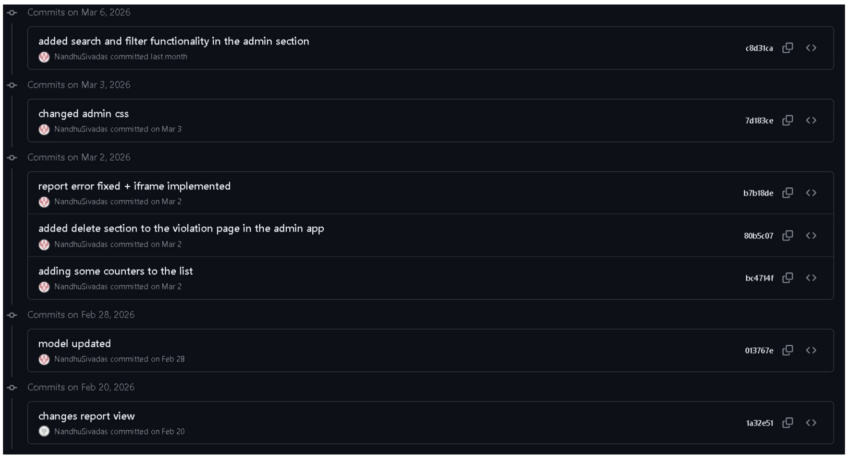

Figure 8.3: Git History

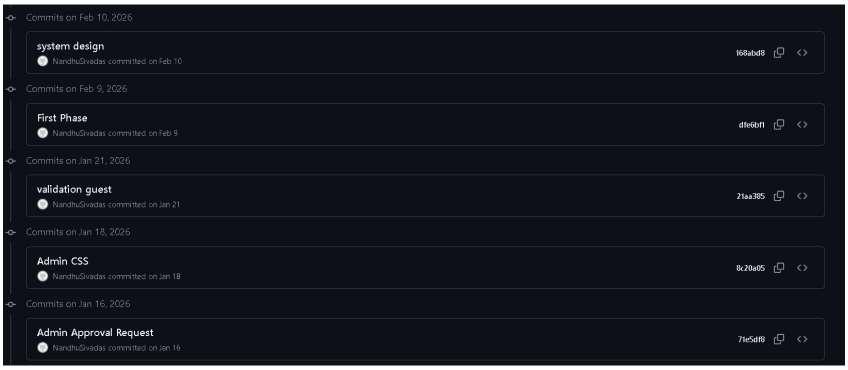

Figure 8.4: Git History

As illustrated in Figures 8.1 through 8.4, the repository timeline reflects key milestones, starting from the initial system design and base setup in early February 2026. Significant progress is visible through commits regarding the development of specific modules, such as Admin CSS styling, Guest validation logic, and the integration of the core AI model updates. The logs further show iterative improvements to the user interface, including the styling of various apps and the addition of search and filter functionalities in the admin section to enhance data management.

The commit history also highlights critical backend developments, such as the implementation of database diagrams, the configuration of the SQLite database, and the final deployment tests on the AWS EC2 instance. Recent updates show refined functionality in the user and report sections, alongside fixes for iframe components and the inclusion of violation counters. Ultimately, this Git history serves as a transparent audit trail, demonstrating consistent progress and rigorous testing throughout the transition from a local prototype to a production-ready cloud application.

9 CONCLUSION

AI-Based Hotel Kitchen Cleanliness Monitoring System marks a significant step toward modernizing hygiene surveillance in the hospitality industry by transitioning from labor-intensive manual inspections to a digital, data-driven framework. Through systematic development using the YOLOv8 architecture, the system provides a comprehensive solution for automating staff attire compliance and pest detection, which enhances overall safety and efficiency within food preparation environments. By providing administrators with real-time access to diagnostic results and automated violation reporting, the platform reduces human error and ensures higher standards of hygiene accountability. Beyond immediate monitoring, the system provides a robust analytical framework that allows management to track recurring hygiene trends and optimize staff training programs.

The integration of a user-friendly Django interface and a robust cloud-hosted infrastructure on AWS EC2 allows for seamless media uploads and instant hygiene predictions, empowering management to make proactive, informed decisions. Features such as automated image analysis, digital record storage, and the generation of official compliance certificates via the ReportLab engine streamline kitchen management and improve public health control. By addressing the inefficiencies of traditional oversight, this system serves as an essential tool for modern food safety, ensuring better resource utilization and a transparent audit trail for the entire hospitality sector. The utilization of an anchor-free detection mechanism within the YOLOv8 architecture ensures that even small objects, such as hairnets or biological hazards, are localized with high precision against complex kitchen backgrounds.

Ultimately, the successful benchmarking of lightweight YOLO models proves that high-precision monitoring can be achieved with minimal computational overhead, making it viable for real-world IoT deployment. This research further validates the feasibility of deploying nano-scale models on cloud-based infrastructure, proving that high-performance AI does not require expensive, high-end hardware. By integrating NMS-free training strategies, the system effectively reduces computational overhead, which is critical for maintaining high performance during peak restaurant operating hours. The project also addresses ethical considerations by advocating for automated safety designs that prioritize public health without infringing upon worker privacy.

The transition to this automated system not only saves time but also ensures a seamless and reliable monitoring experience that meets the rigorous demands of the modern food service industry. Future iterations of this system could incorporate multi-modal IoT sensors to monitor environmental hazards like temperature and gas leaks alongside visual attire violations. This project establishes a scalable foundation for digital health audits that can be easily adopted by regulatory bodies to standardize safety across the global hospitality sector.

10 FUTURE WORK

- Real-Time Video Streaming Integration: Transitioning from static image/video up loads to live CCTV camera feed integration. This will allow for continuous, real-time hygiene surveillance and immediate detection of violations as they occur, ensuring a proactive approach to kitchen safety.
- AutomatedReal-TimeAlerts:2. Implementing a notification system using SMS, Email, or WhatsApp to instantly inform hotel management and administrators when a critical violation, such as a pest sighting or a major hygiene breach, is detected. This will signif icantly reduce response times and ensure better compliance with hygiene standards.
- IoT Sensor Fusion: Enhancing the vision-based system by integrating IoT sensors to monitor environmental factors such as kitchen temperature, humidity, and gas leakage. Combining visual data with environmental metrics will provide a more comprehensive and intelligent analysis of kitchen safety and hygiene management.
- Advanced Behavioral Analysis:Developing deep learning models to recognize specific human activities and food handling practices beyond simple attire compliance. This would involve detecting improper cross-contamination behaviors or unsafe cooking prac tices to provide a holistic view of personnel hygiene.
- Multi-Model Benchmarking:Expanding the architecture to support and compare mul tiple lightweight models like YOLOv10 or YOLOv12 to optimize performance for dif ferent hardware environments. This ensures the system remains efficient for deployment on mobile devices or edge computing hardware.
- Predictive Hygiene Analytics:Utilizing historical data stored in the SQLite database to identify trends and patterns in hygiene violations. By applying machine learning to past records, the system could predict potential ”high-risk” periods and suggest preventive measures to hotel management.

11 APPENDIX

11.1 Minimum Software Requirements

The AI-Based Hotel Kitchen Cleanliness Monitoring System is a web-based application de signed for cross-platform compatibility. To ensure optimal performance, the following software environment is recommended:

Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest versions).
Operating System	Supports Windows, macOS, and Linux platforms.
Internet Connection	A stable connection is required for media uploads and report generation.
Security Permissions	Cookies and JavaScript must be enabled for full functionality.

11.2 Minimum Hardware Requirements

Processor(CPU)	Dual-core or higher (e.g., Intel Core i3 or equivalent).
RAM	At least 4 GB of RAM is recommended for browser-based operations.
Internet Connection	A stable connection is required for media uploads and report generation.
Storage	Approximately 500 MB of free disk space for cache and PDF storage.

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